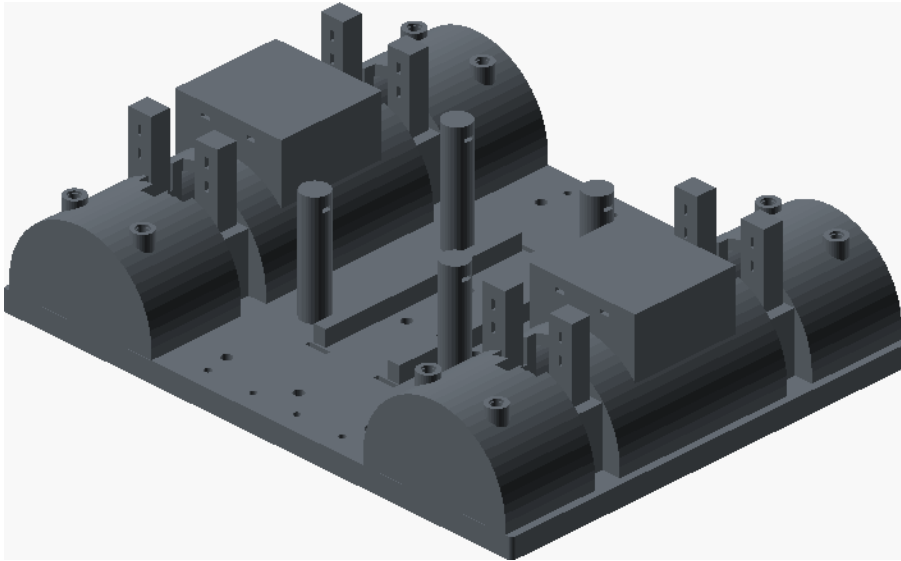


# WOPR: ties, sensors and lighting

5 mm structural walls. 10 mm bases and motor feet. Screw-cap and zip-tie choices.



Both organizers and all three robots are included.

Continuous side skirts; integral wheel wells, motor mounts and battery shelves.

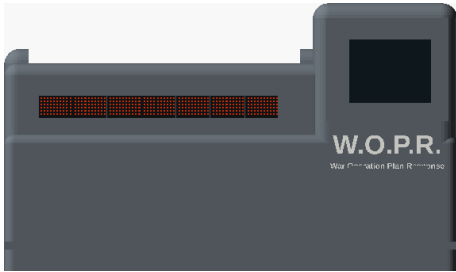
Use matching reinforced parts. Earlier thin-wall exports are historical references.

Fit coupons and assembly checks are supplied; no load, drop or driving test is claimed.

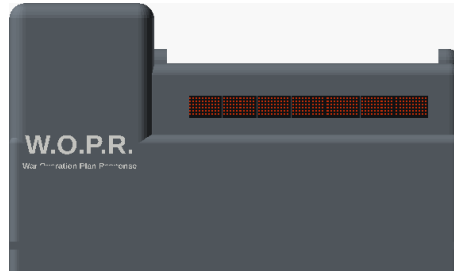
# Seven-per-side organizer

Body 279.4 x 155 x 165 mm; 7 RGB modules per side. Nominal walls 5 mm, base 10 mm.

Front



Rear



Low end



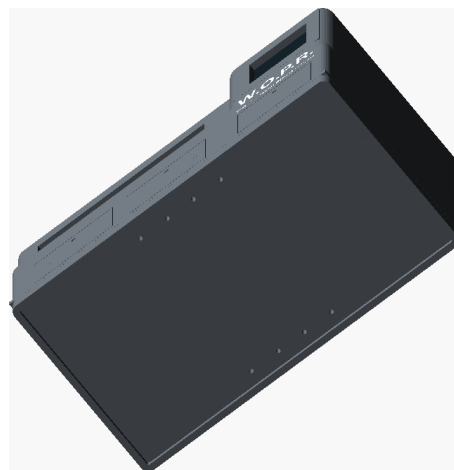
Tower end



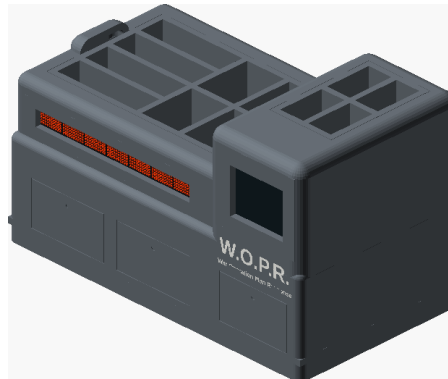
Top oblique



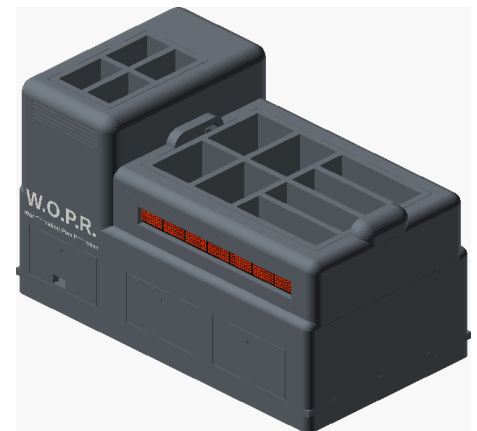
Bottom oblique



Front oblique



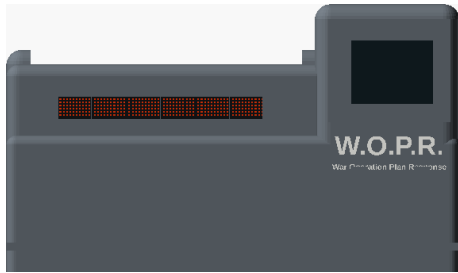
Rear oblique



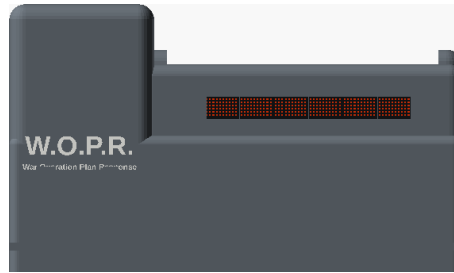
# Six-per-side organizer

Body 279.4 x 155 x 165 mm; 6 RGB modules per side. Nominal walls 5 mm, base 10 mm.

Front



Rear



Low end



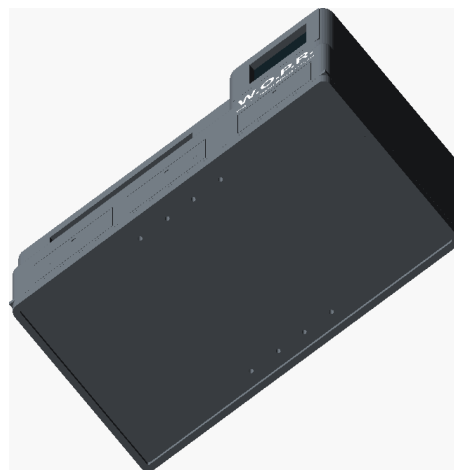
Tower end



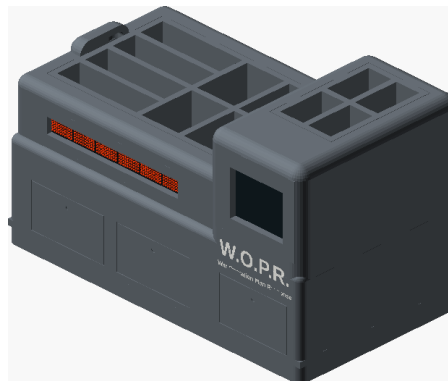
Top oblique



Bottom oblique



Front oblique



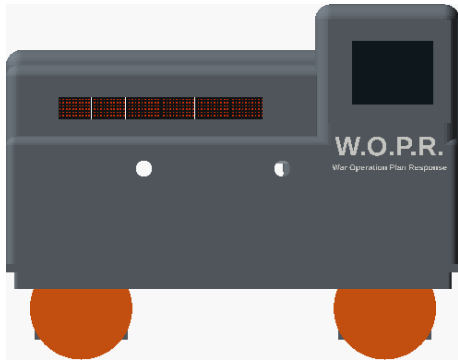
Rear oblique



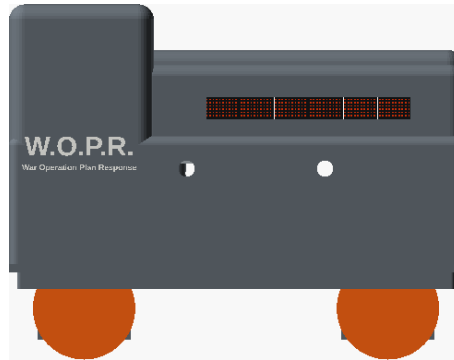
# Wide low-axle eight-wheel robot

Body 279.4 x 220 x 165 mm; 6 RGB modules per side. Nominal walls 5 mm, base 10 mm.

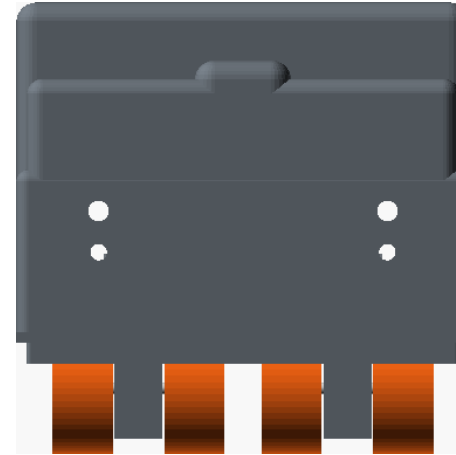
Front



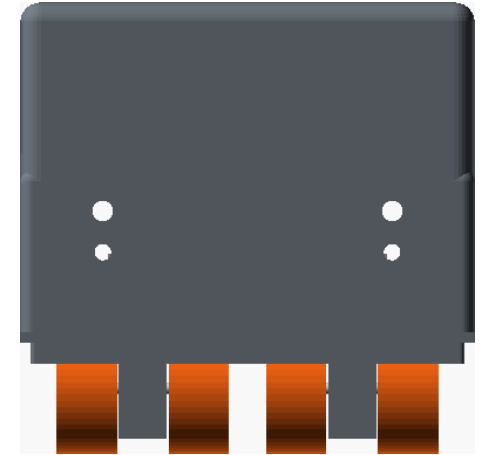
Rear



Low end



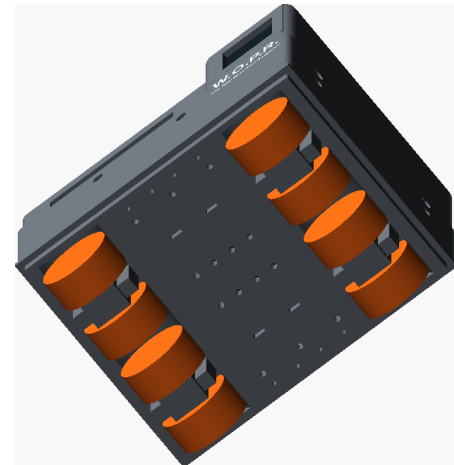
Tower end



Top oblique



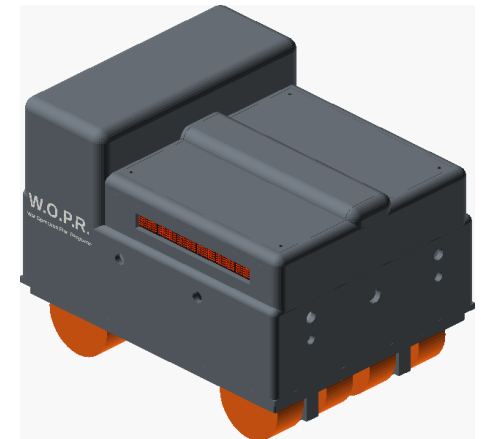
Bottom oblique



Front oblique



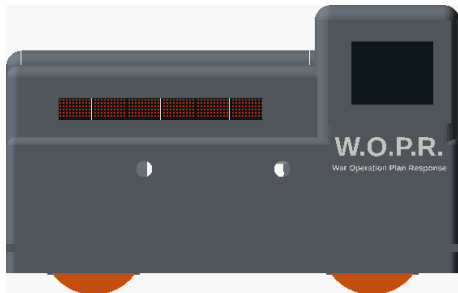
Rear oblique



# Tucked four-wheel robot

Body 279.4 x 158 x 165 mm; 6 RGB modules per side. Nominal walls 5 mm, base 10 mm.

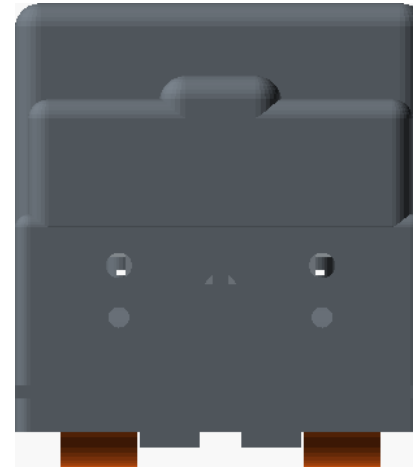
Front



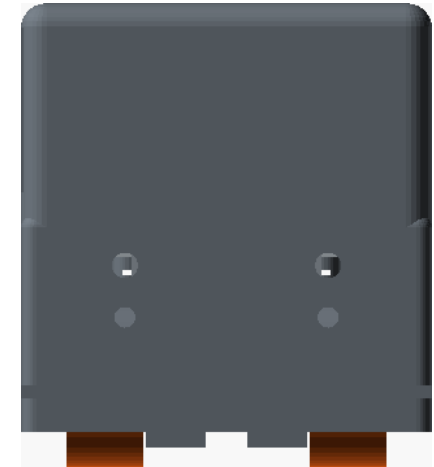
Rear



Low end



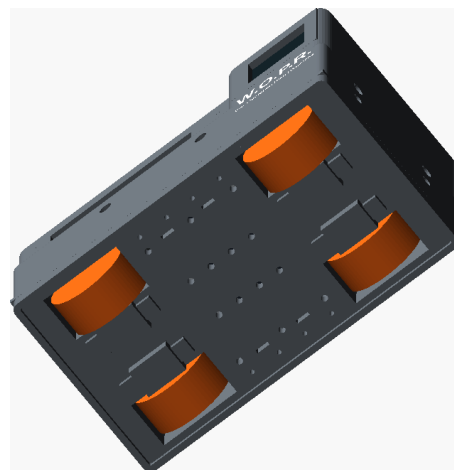
Tower end



Top oblique



Bottom oblique



Front oblique



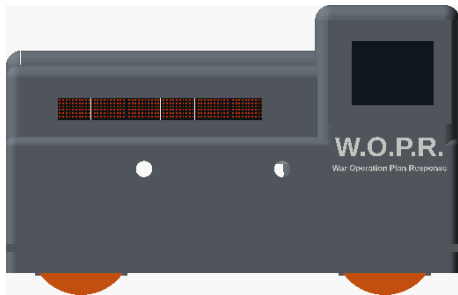
Rear oblique



# Wide tucked eight-wheel robot

Body 279.4 x 220 x 165 mm; 6 RGB modules per side. Nominal walls 5 mm, base 10 mm.

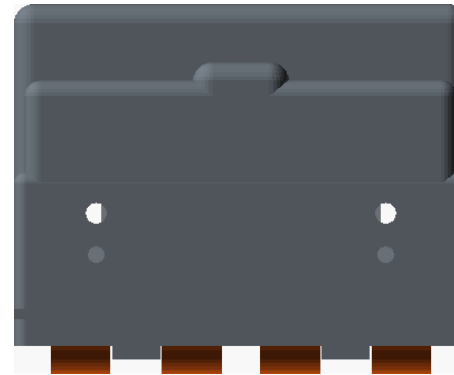
Front



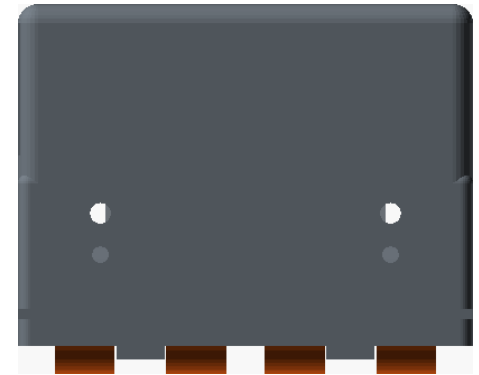
Rear



Low end



Tower end



Top oblique



Bottom oblique



Front oblique

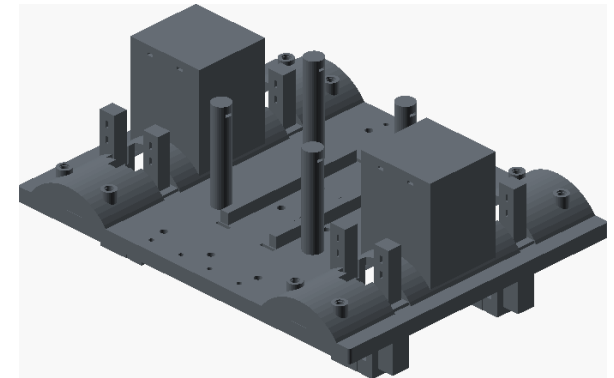
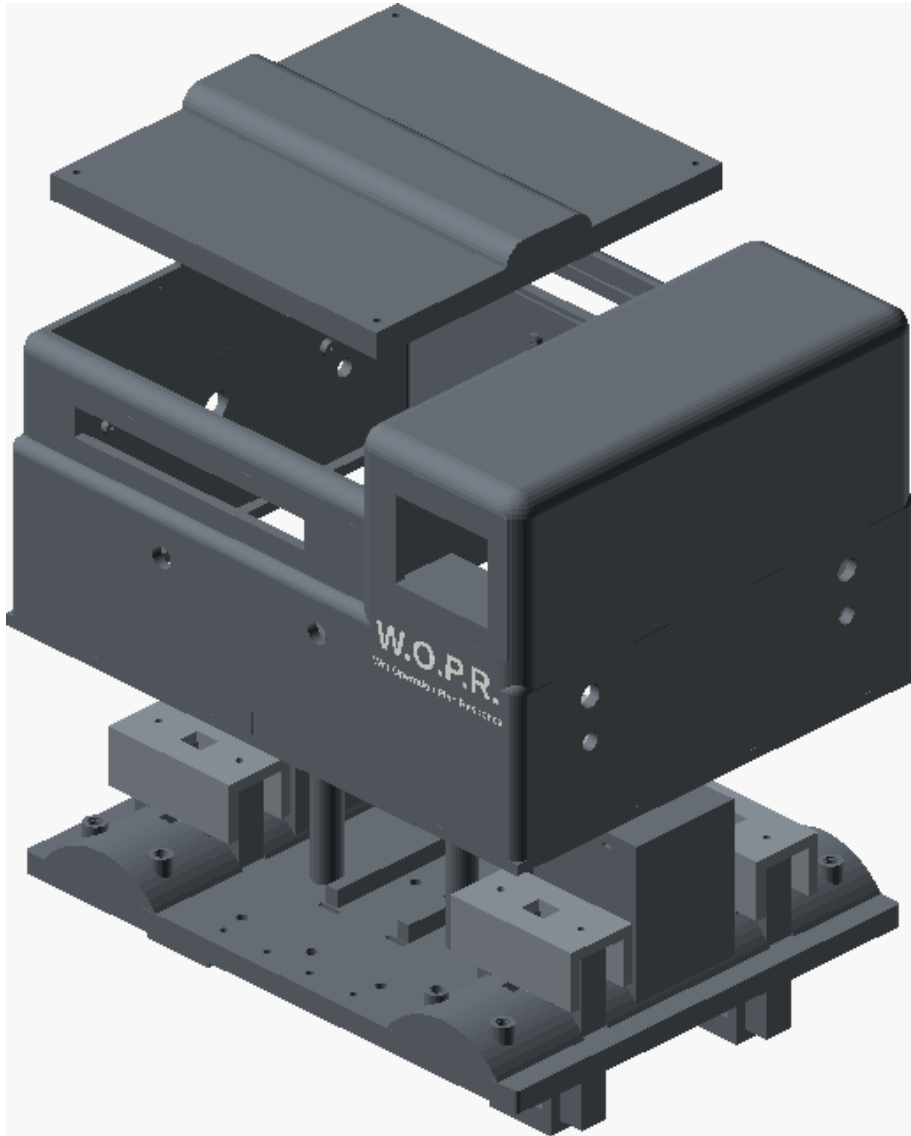


Rear oblique



# Wide low-axe eight-wheel robot

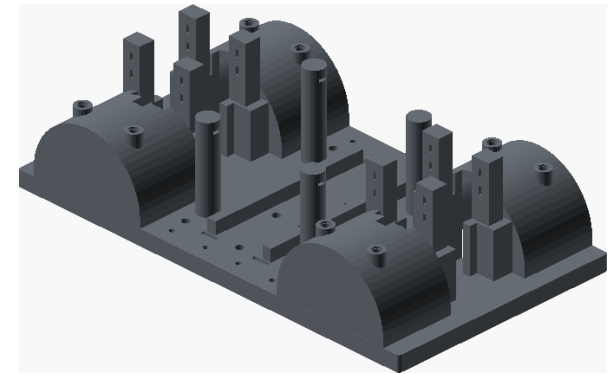
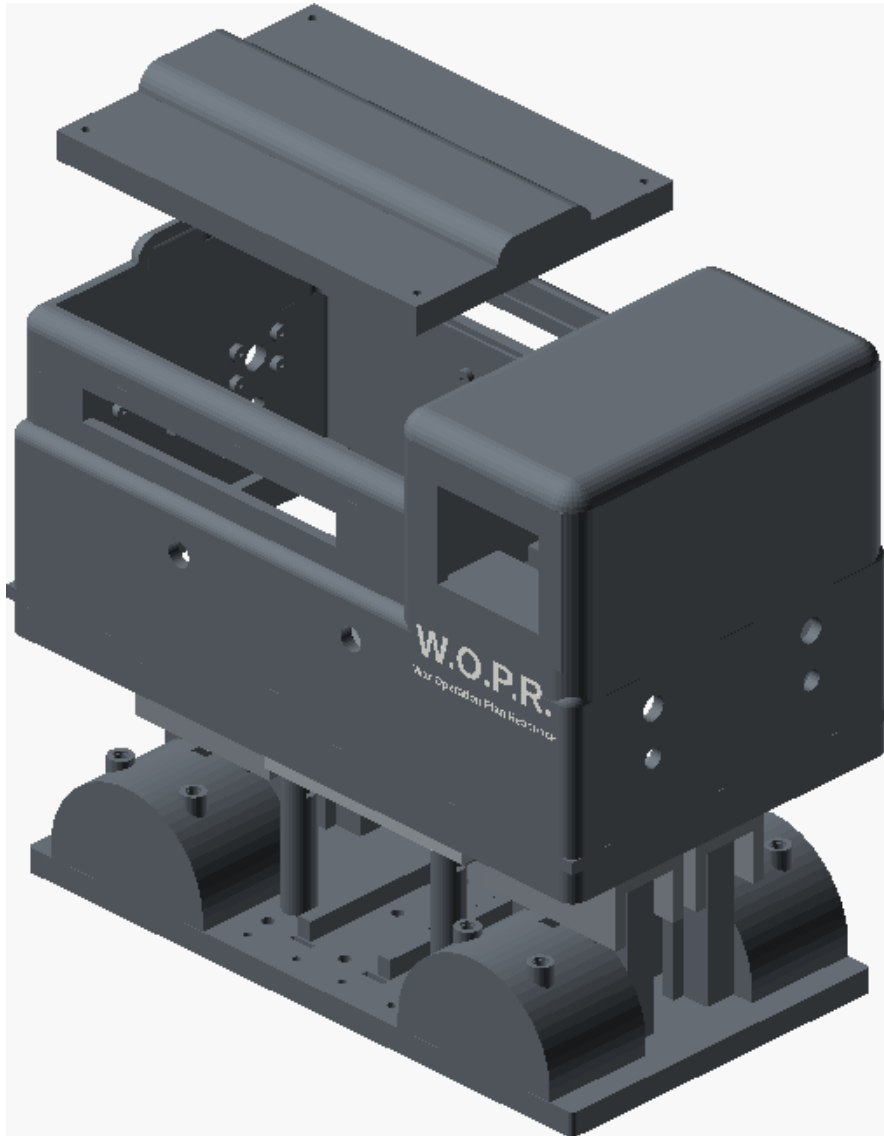
Shell and lid lifted. Motor mounts and wheel wells are integral with the single-piece chassis.



- One structural chassis
- Integral mounts, wells and shelves
- 10 mm base and motor feet
- 5 mm case and wheel-well walls
- Bare motors fit from above
- Wheels fit below, then slide inward
- Three motor ties; tied controller tray

# Tucked four-wheel robot

Shell and lid lifted. Motor mounts and wheel wells are integral with the single-piece chassis.



One structural chassis

Integral mounts, wells and shelves

10 mm base and motor feet

5 mm case and wheel-well walls

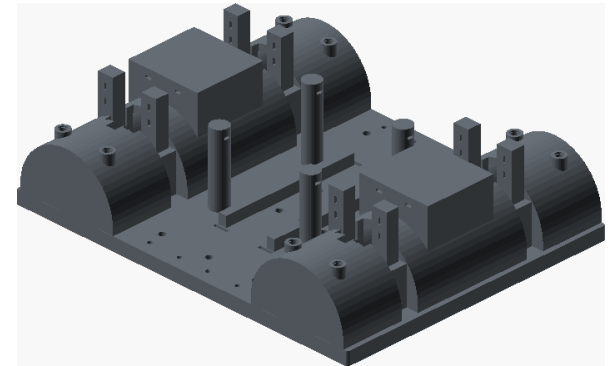
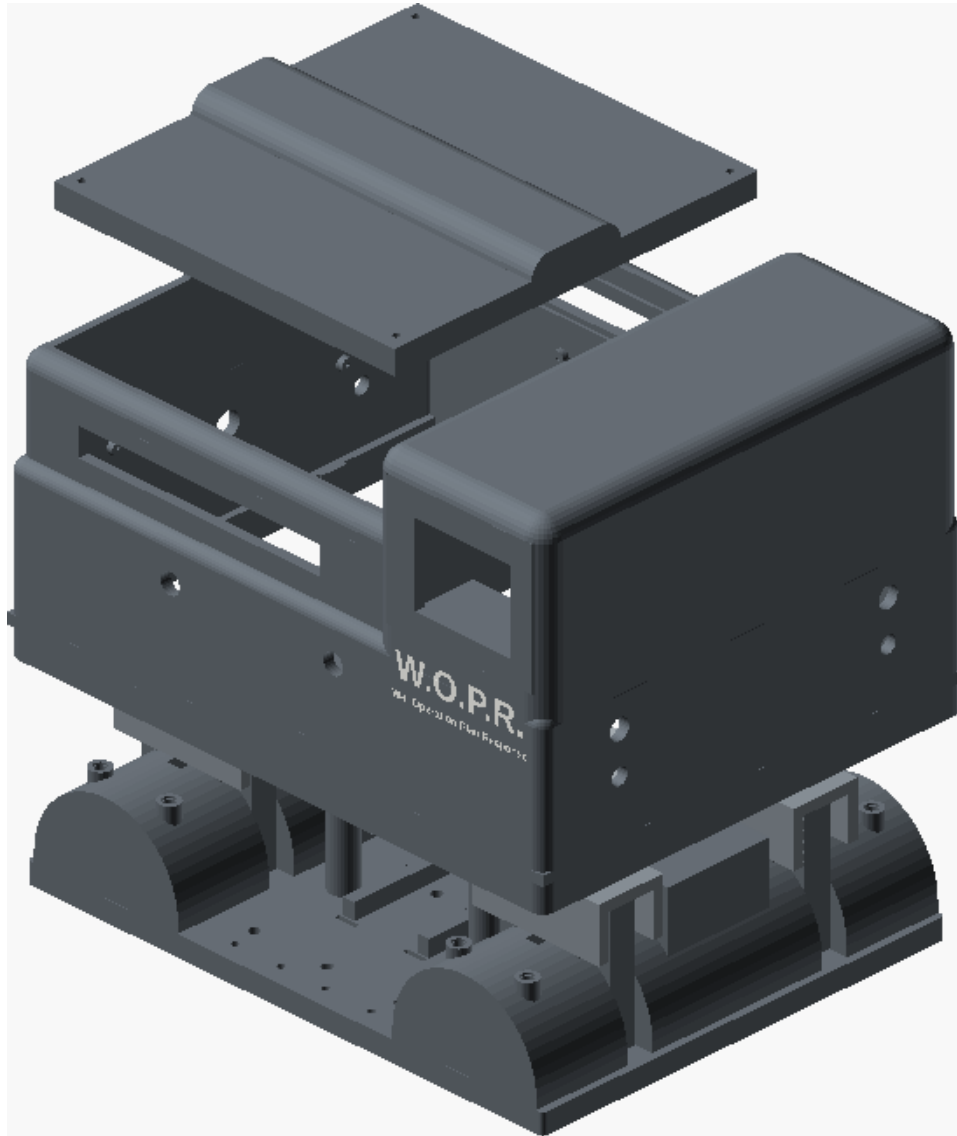
Bare motors fit from above

Wheels fit below, then slide inward

Three motor ties; tied controller tray

# Wide tucked eight-wheel robot

Shell and lid lifted. Motor mounts and wheel wells are integral with the single-piece chassis.



One structural chassis

Integral mounts, wells and shelves

10 mm base and motor feet

5 mm case and wheel-well walls

Bare motors fit from above

Wheels fit below, then slide inward

Three motor ties; tied controller tray

# r-shell

Bounds: 279.40 x 155.00 x 164.95 mm. Actual checked mesh; top/bottom tilted for depth.

Front



Rear



Low end



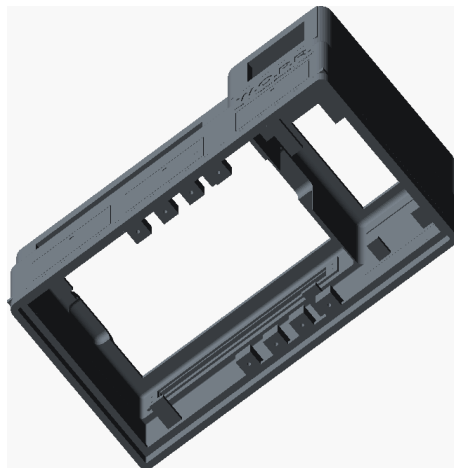
Tower end



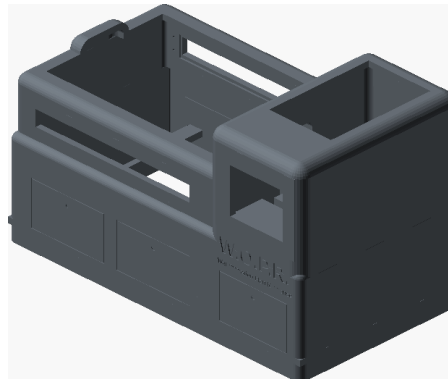
Top oblique



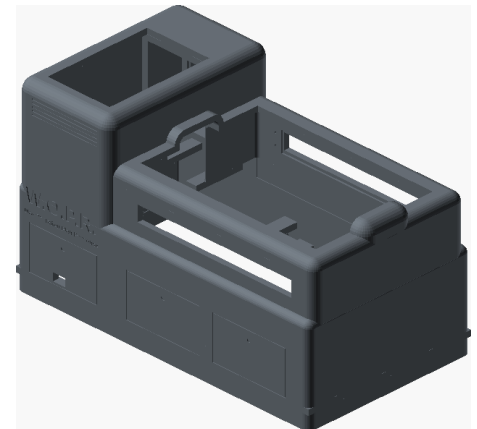
Bottom oblique



Front oblique



Rear oblique



# r-shell

Bounds: 279.40 x 155.00 x 164.95 mm. Actual checked mesh; top/bottom tilted for depth.

Front



Rear



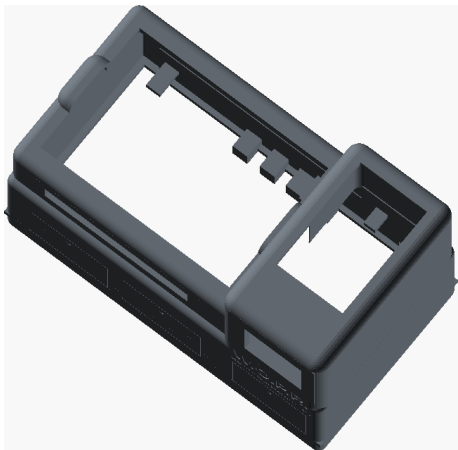
Low end



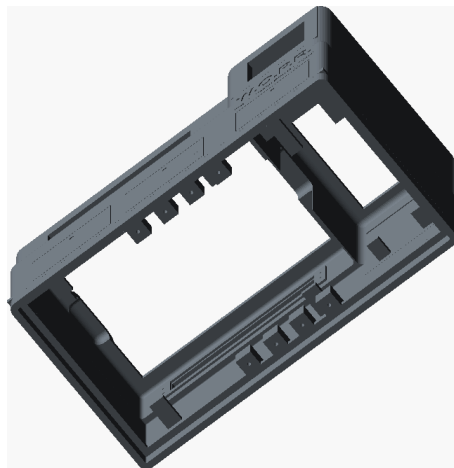
Tower end



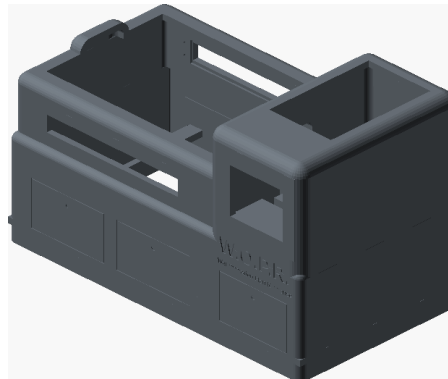
Top oblique



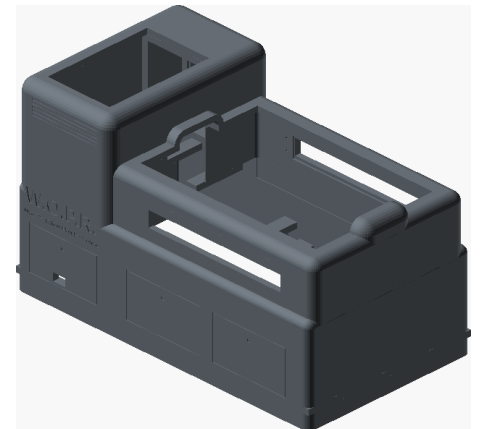
Bottom oblique



Front oblique



Rear oblique



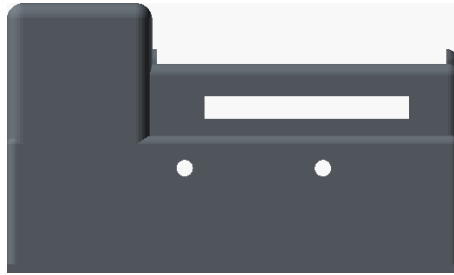
# r-shell

Bounds: 279.40 x 220.00 x 164.95 mm. Actual checked mesh; top/bottom tilted for depth.

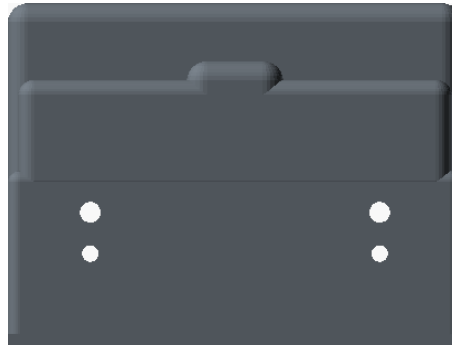
Front



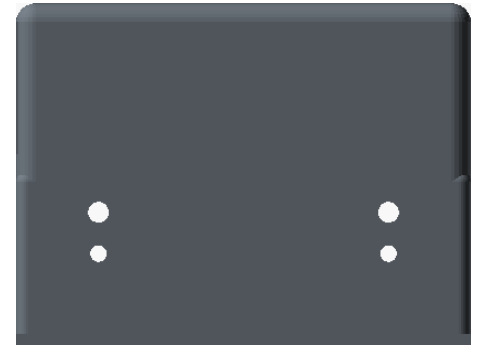
Rear



Low end



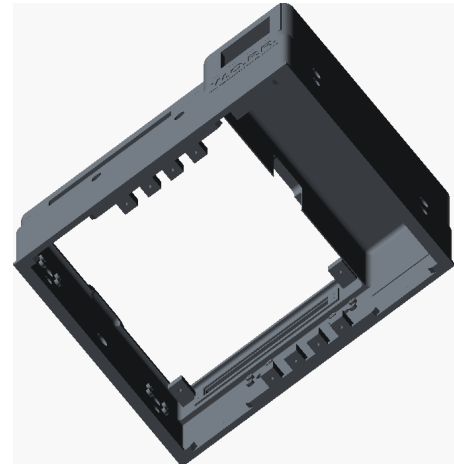
Tower end



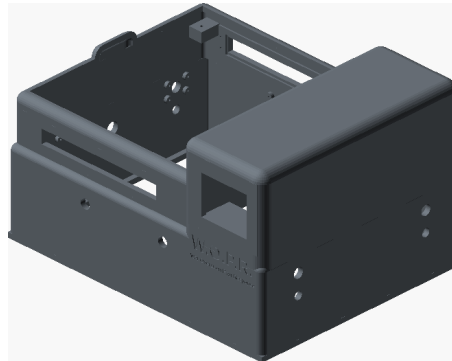
Top oblique



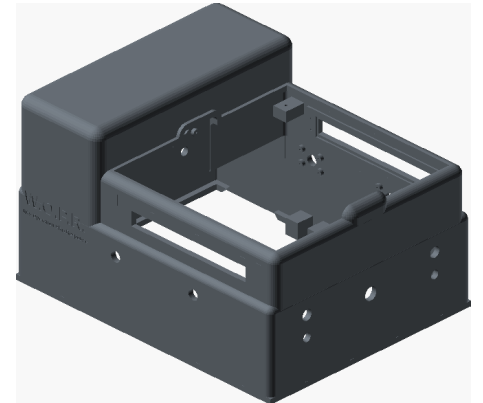
Bottom oblique



Front oblique



Rear oblique



# r-shell

Bounds: 279.40 x 158.00 x 164.95 mm. Actual checked mesh; top/bottom tilted for depth.

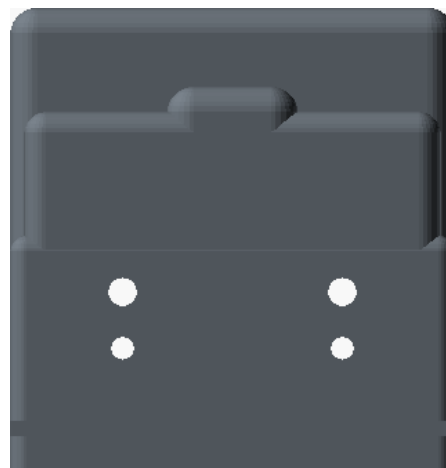
Front



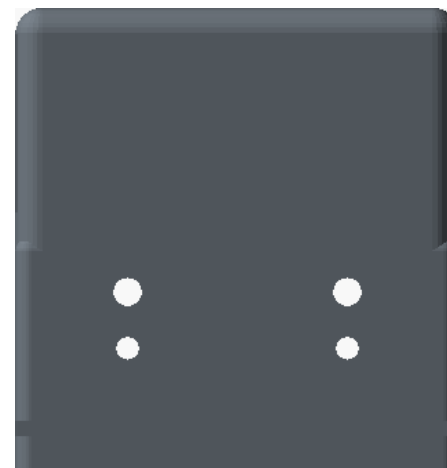
Rear



Low end



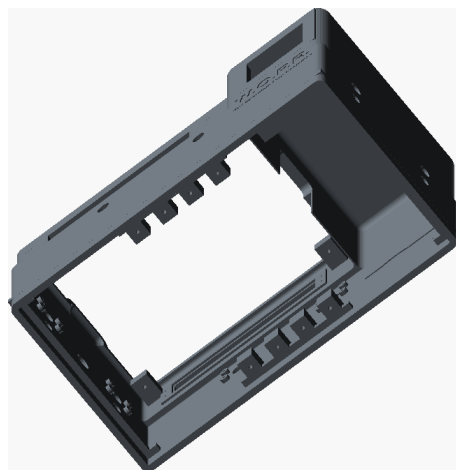
Tower end



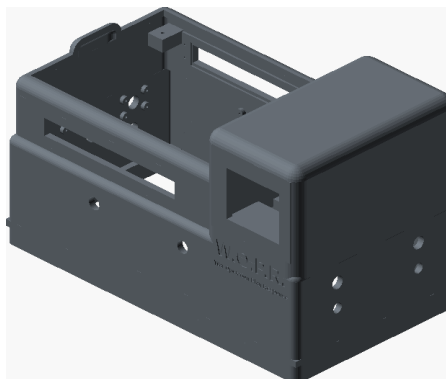
Top oblique



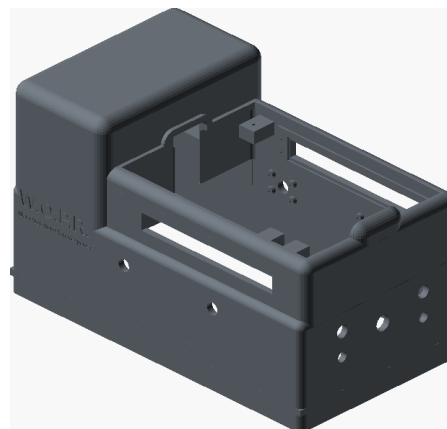
Bottom oblique



Front oblique



Rear oblique



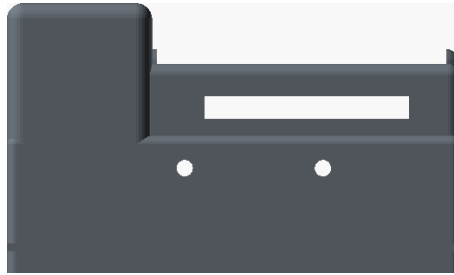
# r-shell

Bounds: 279.40 x 220.00 x 164.95 mm. Actual checked mesh; top/bottom tilted for depth.

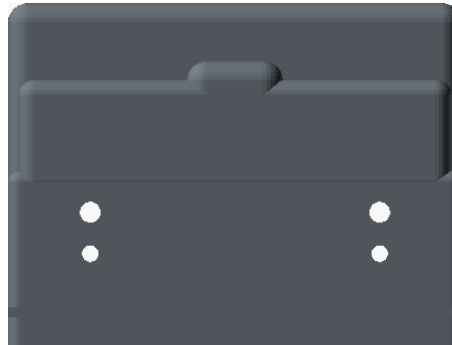
Front



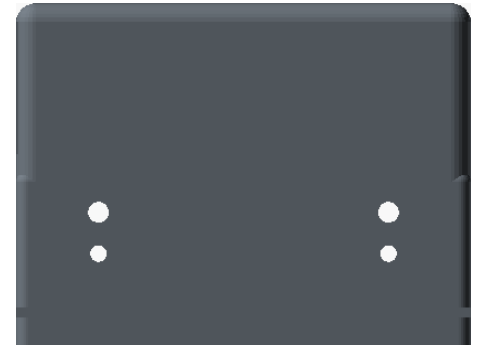
Rear



Low end



Tower end



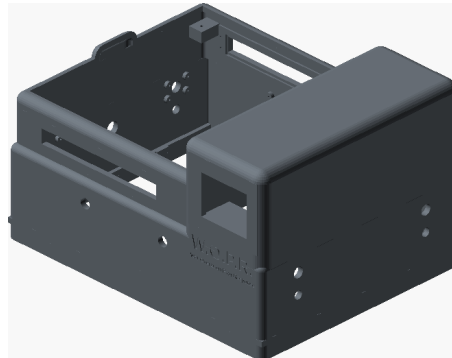
Top oblique



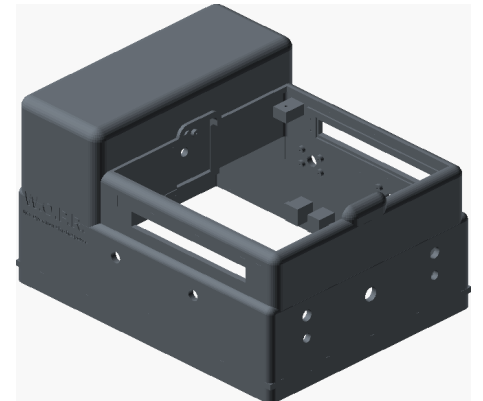
Bottom oblique



Front oblique



Rear oblique



# r-frame

Bounds: 268.80 x 144.40 x 10.00 mm. Actual checked mesh; top/bottom tilted for depth.

Front

Rear

Low end

Tower end

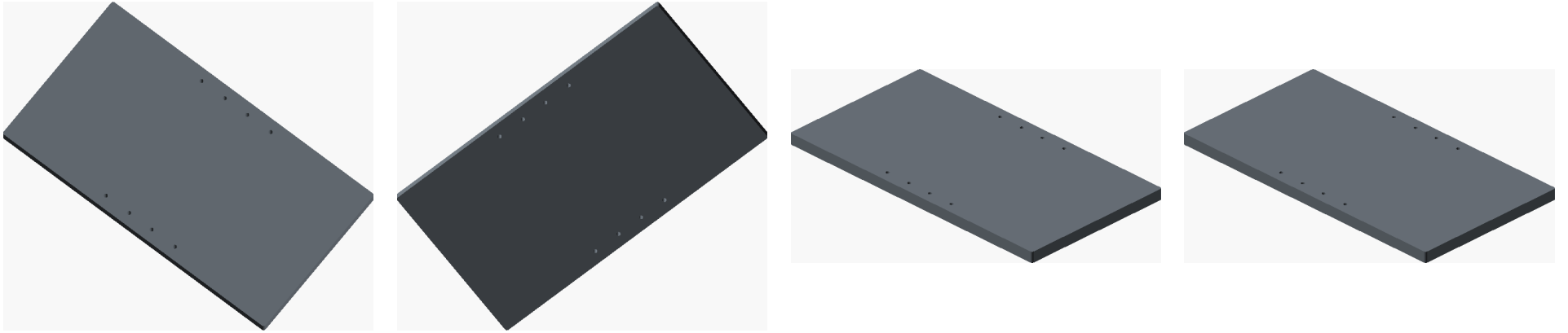


Top oblique

Bottom oblique

Front oblique

Rear oblique



# r-frame

Bounds: 268.80 x 209.60 x 118.70 mm. Actual checked mesh; top/bottom tilted for depth.

Front



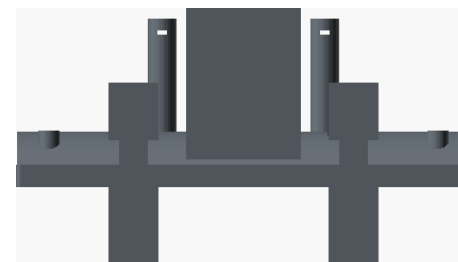
Rear



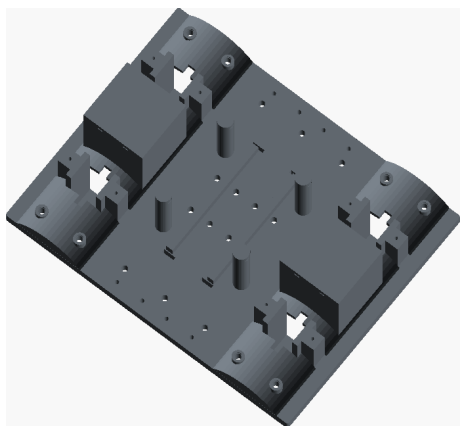
Low end



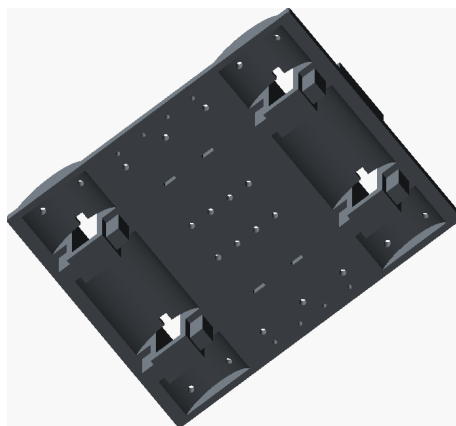
Tower end



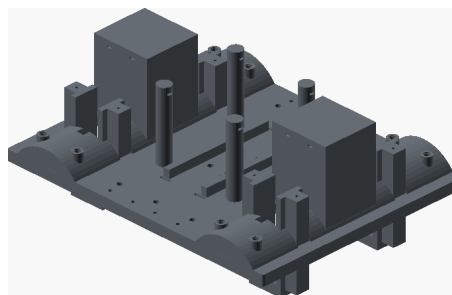
Top oblique



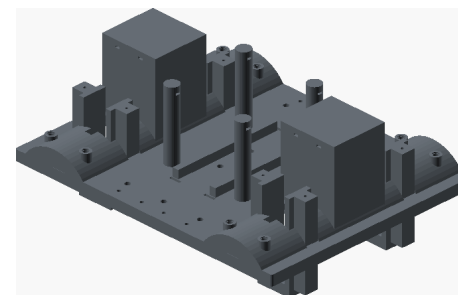
Bottom oblique



Front oblique



Rear oblique



# r-frame

Bounds: 268.80 x 147.60 x 84.40 mm. Actual checked mesh; top/bottom tilted for depth.

Front



Rear



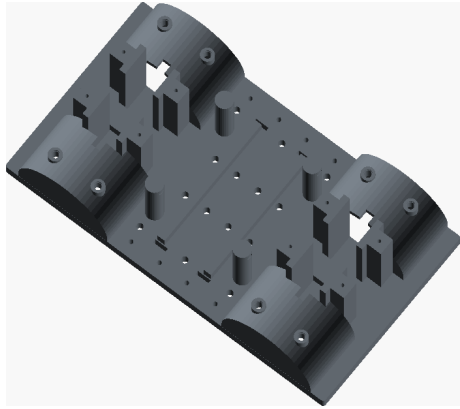
Low end



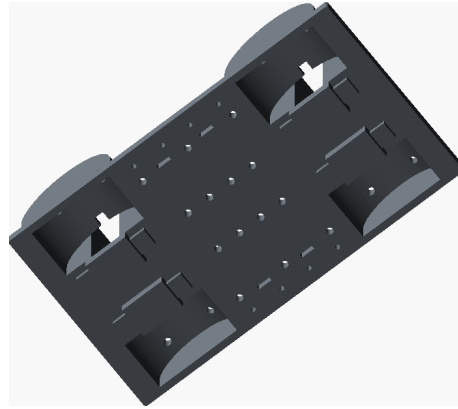
Tower end



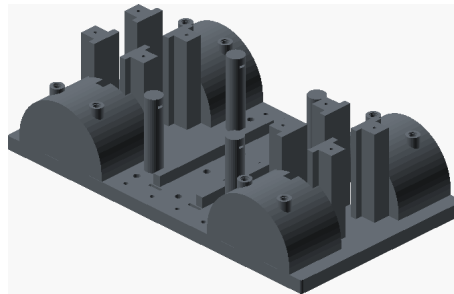
Top oblique



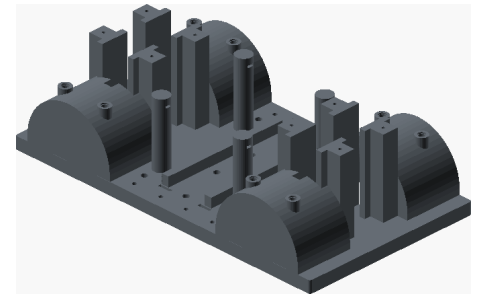
Bottom oblique



Front oblique



Rear oblique



# r-frame

Bounds: 268.80 x 209.60 x 84.40 mm. Actual checked mesh; top/bottom tilted for depth.

Front



Rear



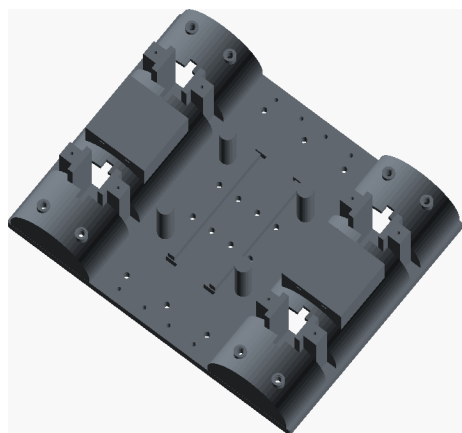
Low end



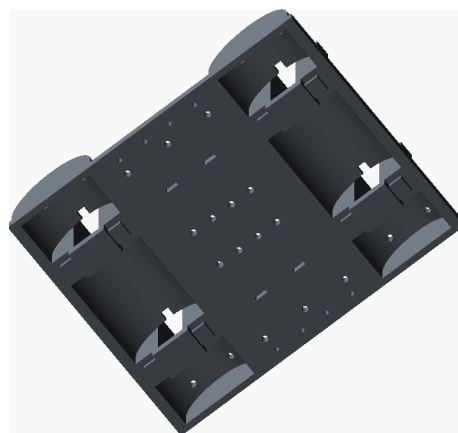
Tower end



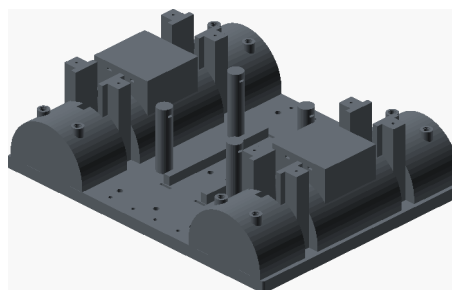
Top oblique



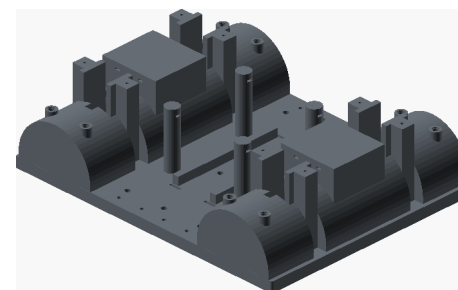
Bottom oblique



Front oblique



Rear oblique



# r-retainer

Bounds: 168.49 x 28.38 x 5.00 mm. Actual checked mesh; top/bottom tilted for depth.

Front

Rear

Low end

Tower end

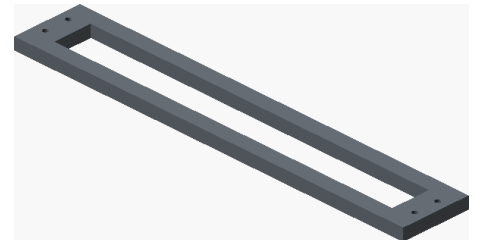
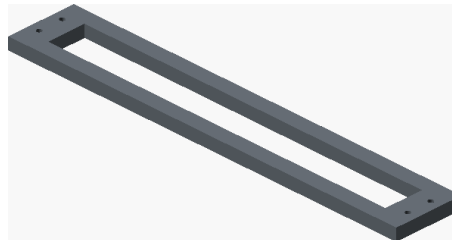
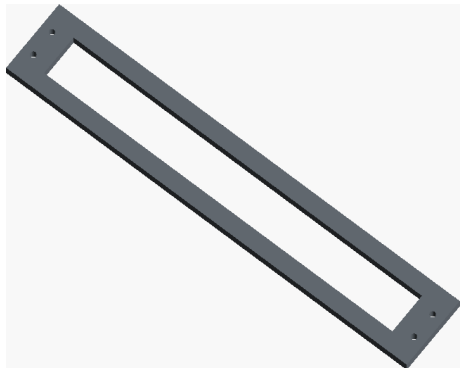


Top oblique

Bottom oblique

Front oblique

Rear oblique



# r-retainer

Bounds: 147.23 x 28.38 x 5.00 mm. Actual checked mesh; top/bottom tilted for depth.

Front



Rear



Low end



Tower end



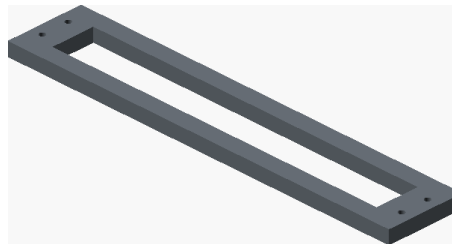
Top oblique



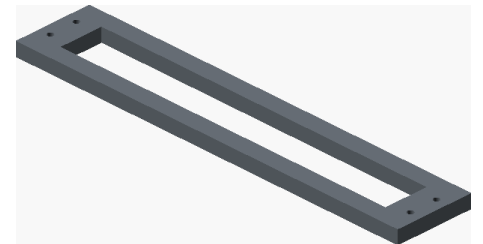
Bottom oblique



Front oblique



Rear oblique



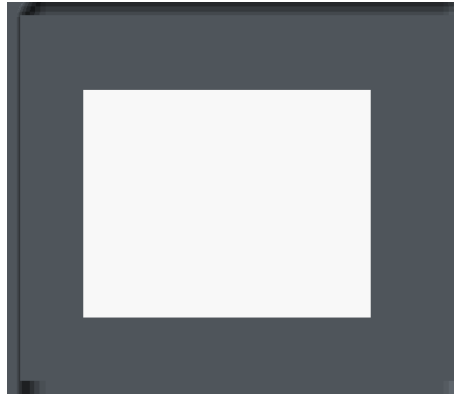
# r-lcd-coupon

Bounds: 79.00 x 42.00 x 68.00 mm. Actual checked mesh; top/bottom tilted for depth.

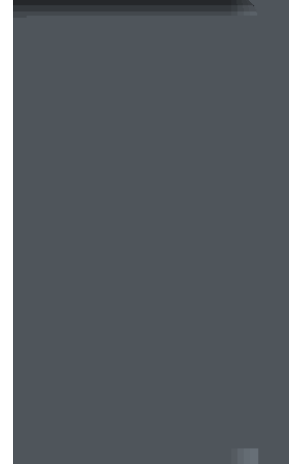
Front



Rear



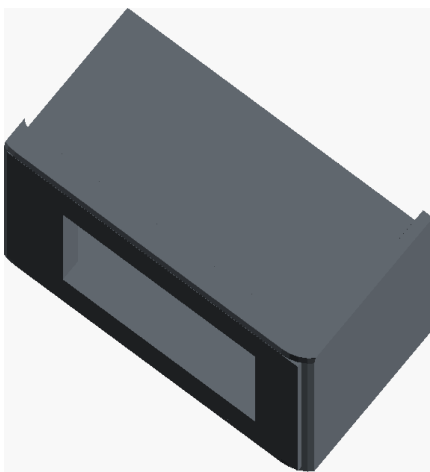
Low end



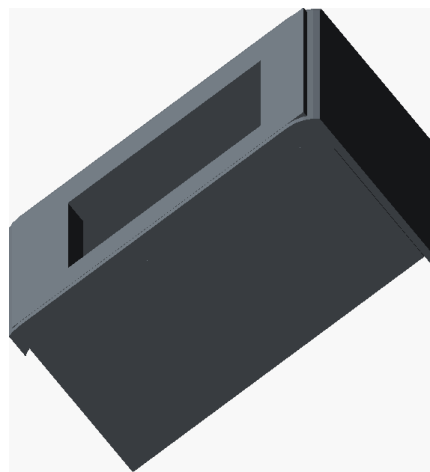
Tower end



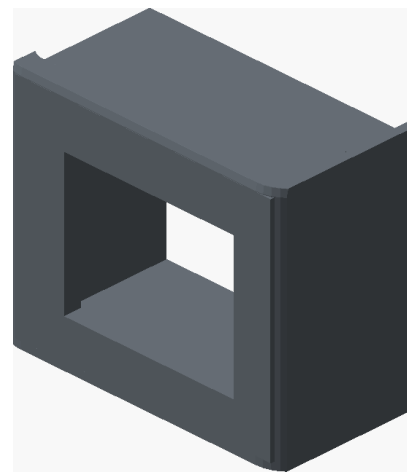
Top oblique



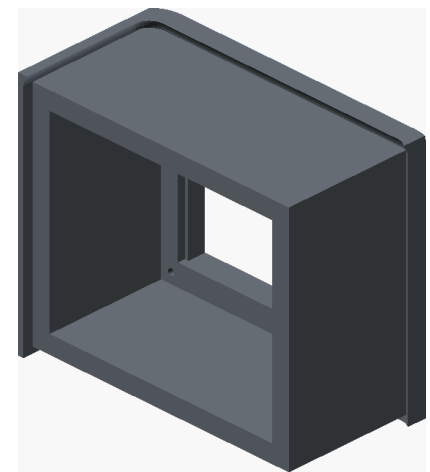
Bottom oblique



Front oblique



Rear oblique



# r-cup-left

Bounds: 166.40 x 111.00 x 88.00 mm. Actual checked mesh; top/bottom tilted for depth.

Front



Rear



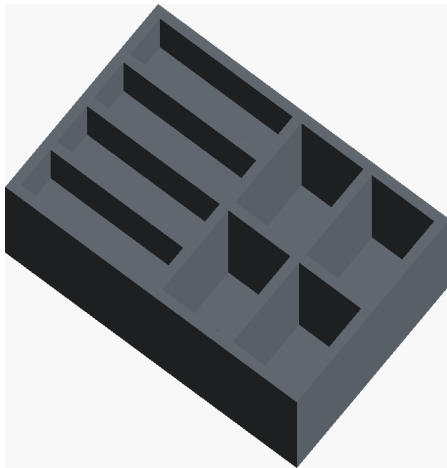
Low end



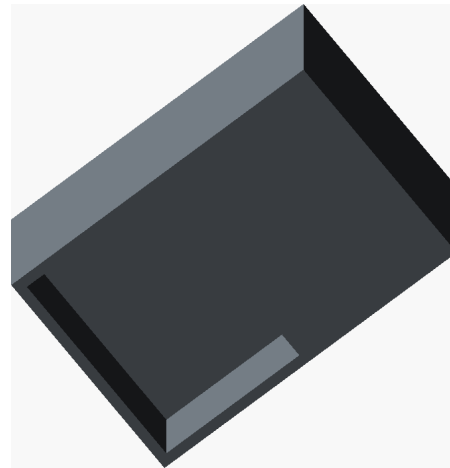
Tower end



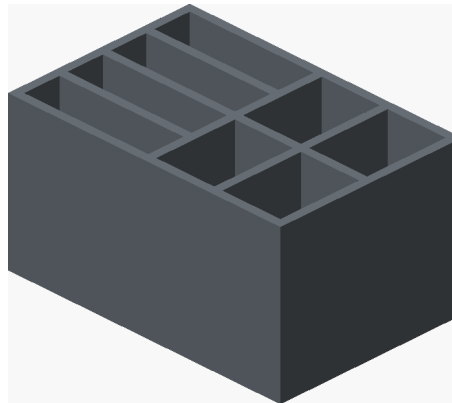
Top oblique



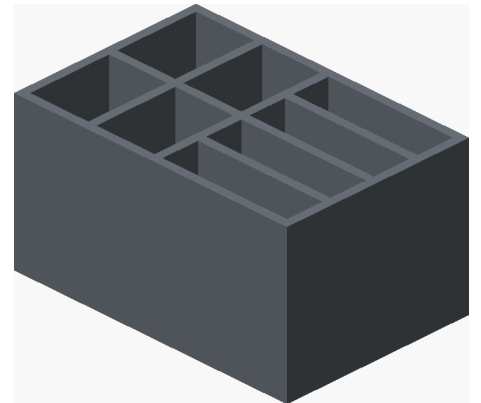
Bottom oblique



Front oblique



Rear oblique



# r-cup-right

Bounds: 166.40 x 111.00 x 88.00 mm. Actual checked mesh; top/bottom tilted for depth.

Front



Rear



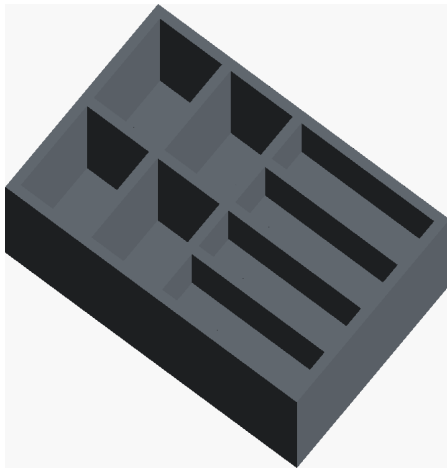
Low end



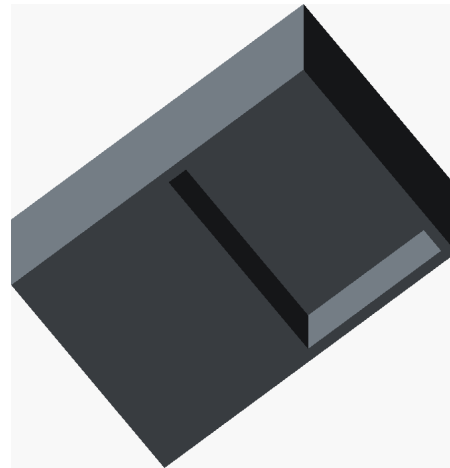
Tower end



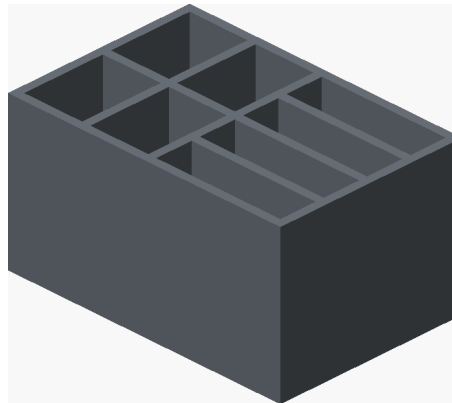
Top oblique



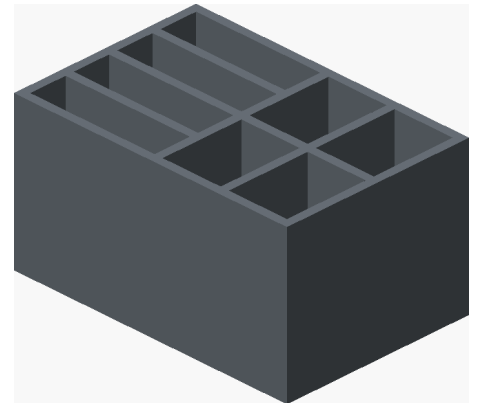
Bottom oblique



Front oblique



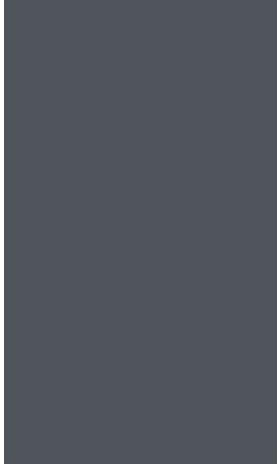
Rear oblique



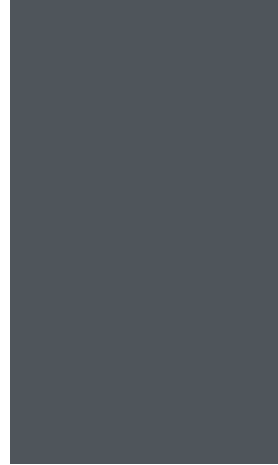
# r-cup-tower

Bounds: 58.40 x 94.40 x 100.00 mm. Actual checked mesh; top/bottom tilted for depth.

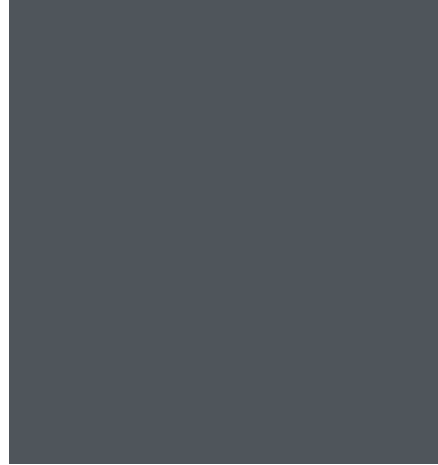
Front



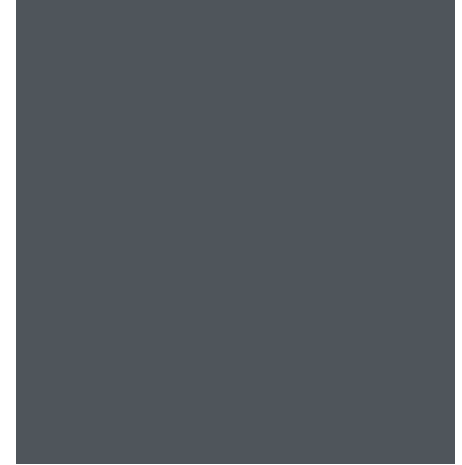
Rear



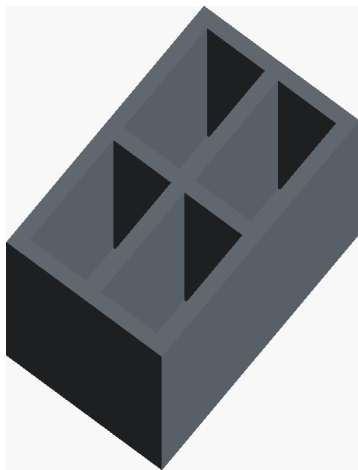
Low end



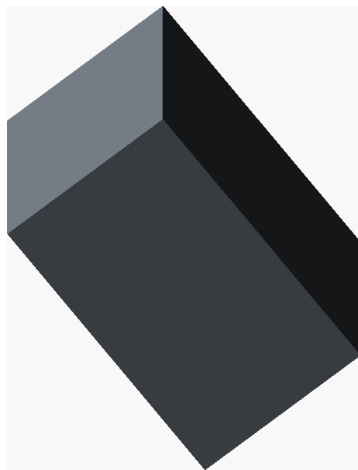
Tower end



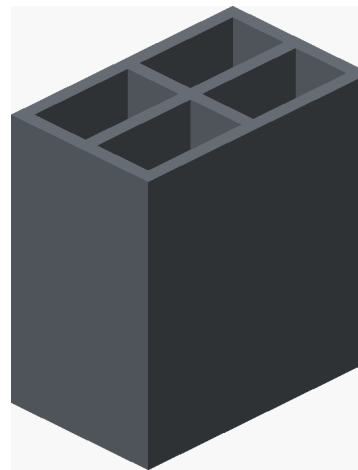
Top oblique



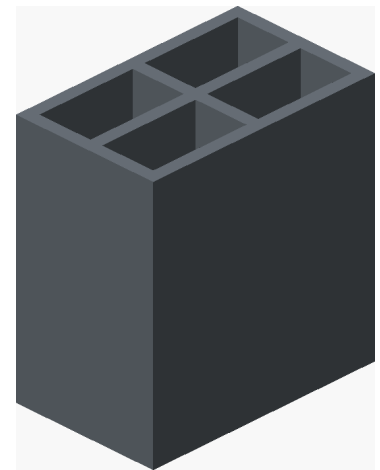
Bottom oblique



Front oblique



Rear oblique



# r-lid

Bounds: 177.40 x 127.40 x 20.65 mm. Actual checked mesh; top/bottom tilted for depth.

Front



Rear



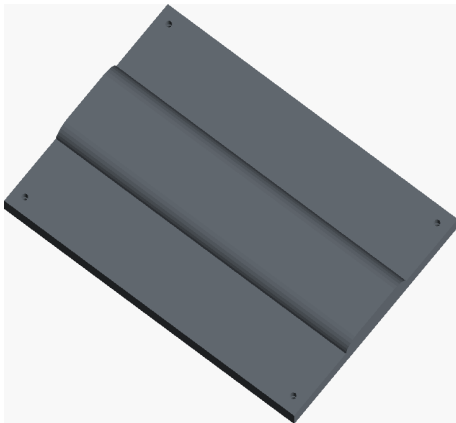
Low end



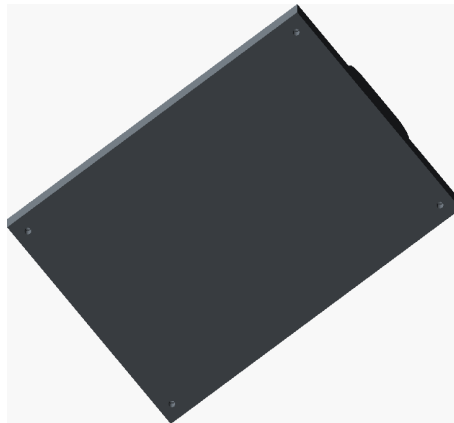
Tower end



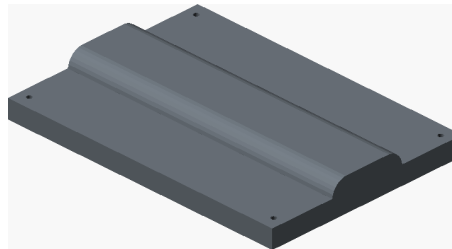
Top oblique



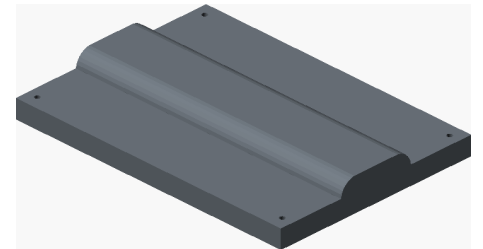
Bottom oblique



Front oblique



Rear oblique



# r-lid

Bounds: 177.40 x 189.40 x 20.65 mm. Actual checked mesh; top/bottom tilted for depth.

Front



Rear



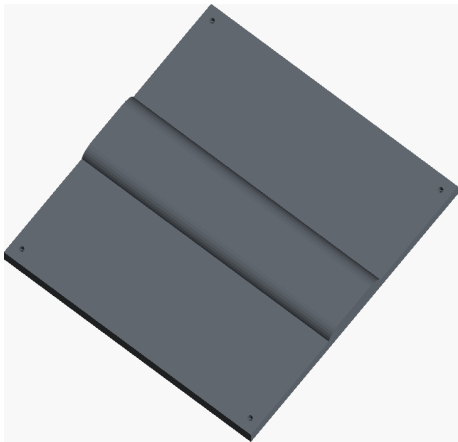
Low end



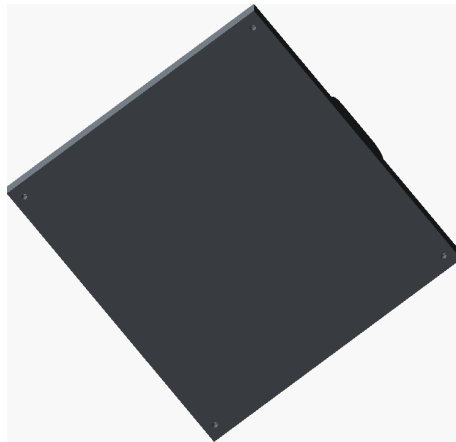
Tower end



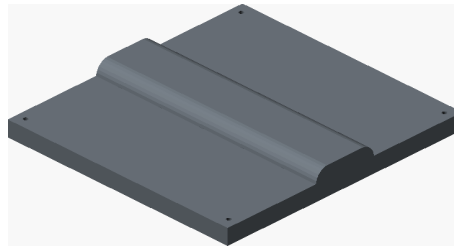
Top oblique



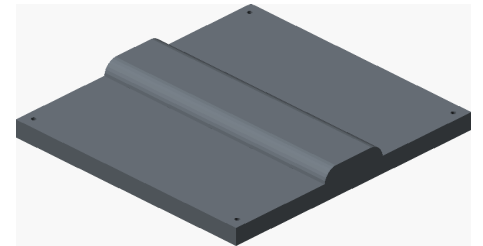
Bottom oblique



Front oblique



Rear oblique



# r-tray

Bounds: 98.00 x 88.00 x 5.00 mm. Actual checked mesh; top/bottom tilted for depth.

Front

Rear

Low end

Tower end

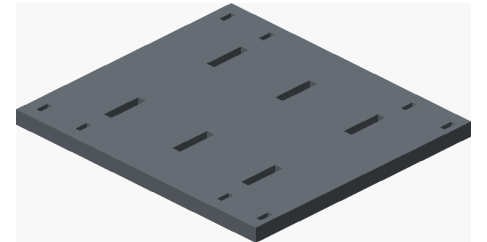
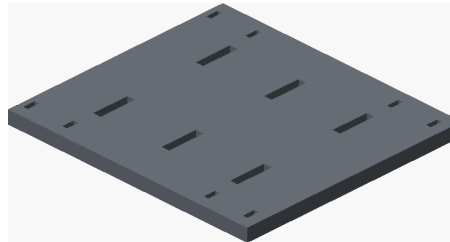
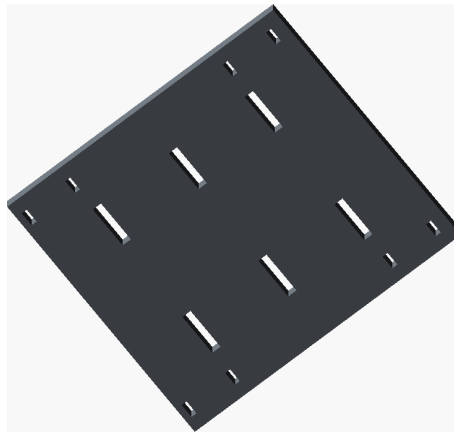
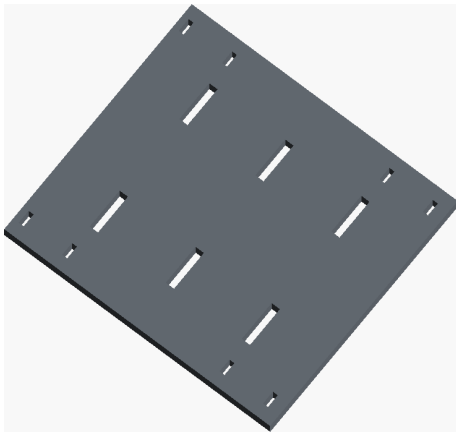


Top oblique

Bottom oblique

Front oblique

Rear oblique



# r-frame-zip

Bounds: 268.80 x 209.60 x 118.70 mm. Actual checked mesh; top/bottom tilted for depth.

Front



Rear



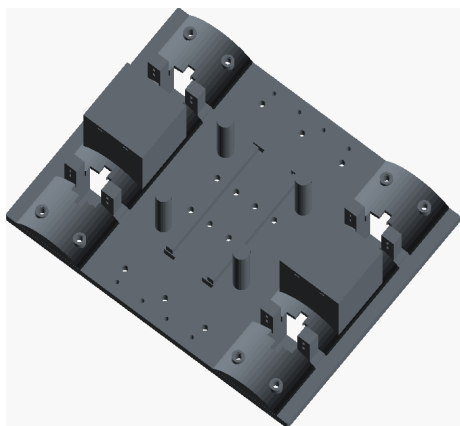
Low end



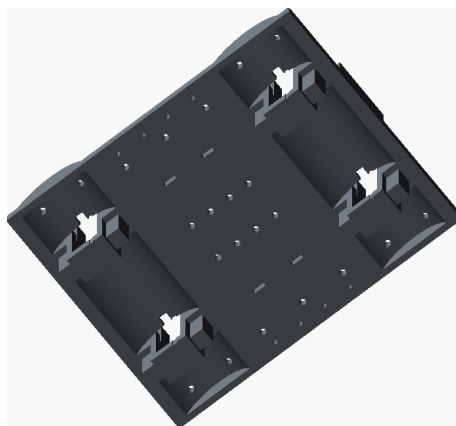
Tower end



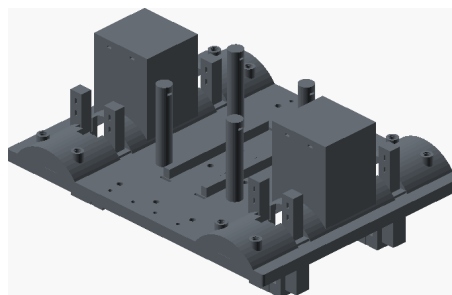
Top oblique



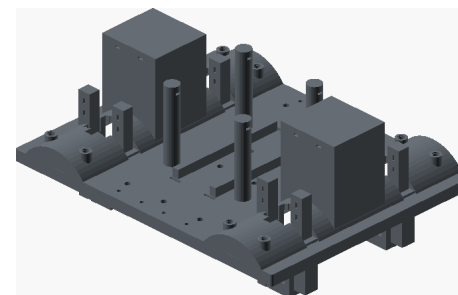
Bottom oblique



Front oblique



Rear oblique



# r-frame-zip

Bounds: 268.80 x 147.60 x 84.40 mm. Actual checked mesh; top/bottom tilted for depth.

Front



Rear



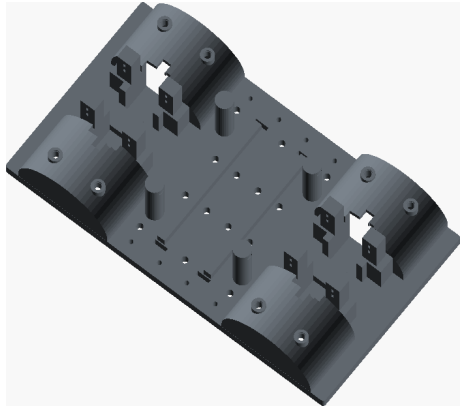
Low end



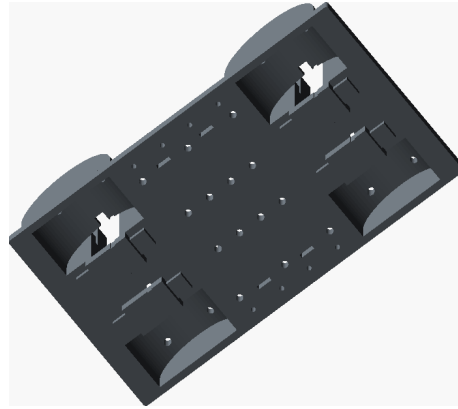
Tower end



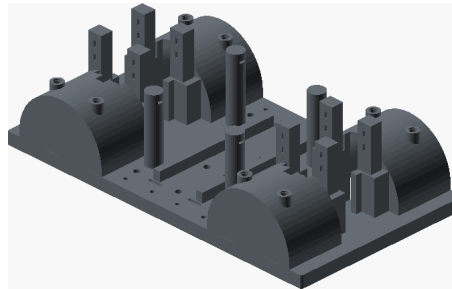
Top oblique



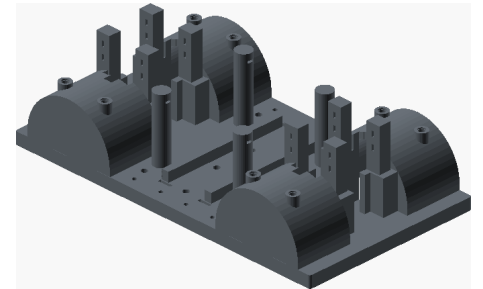
Bottom oblique



Front oblique



Rear oblique



# r-frame-zip

Bounds: 268.80 x 209.60 x 84.40 mm. Actual checked mesh; top/bottom tilted for depth.

Front



Rear



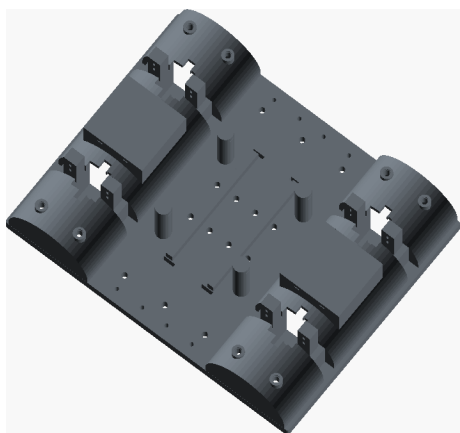
Low end



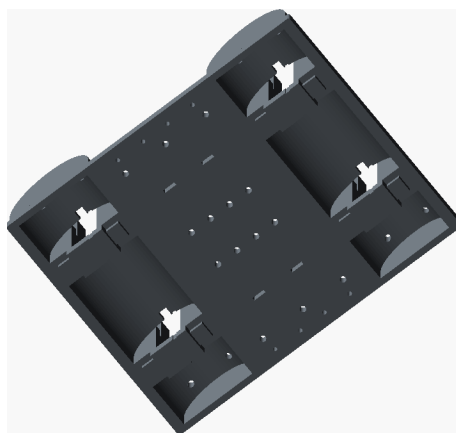
Tower end



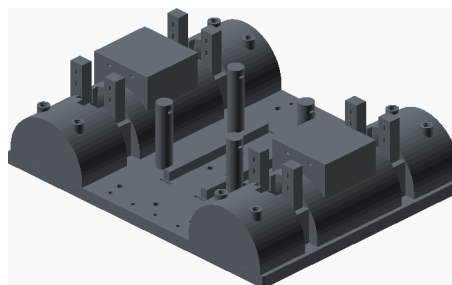
Top oblique



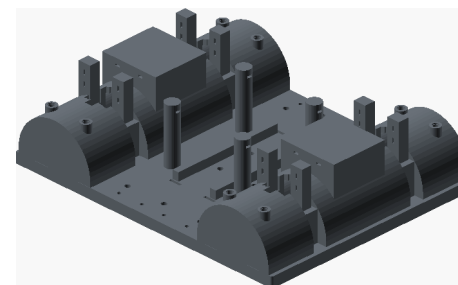
Bottom oblique



Front oblique



Rear oblique



# r-cap

Bounds: 66.00 x 33.40 x 26.30 mm. Actual checked mesh; top/bottom tilted for depth.

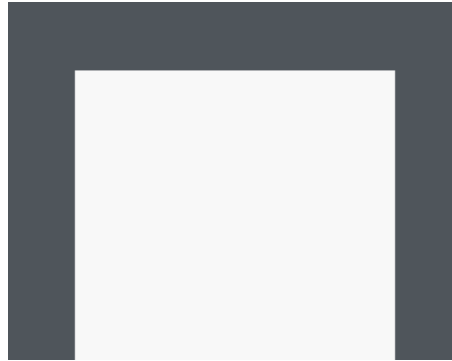
Front



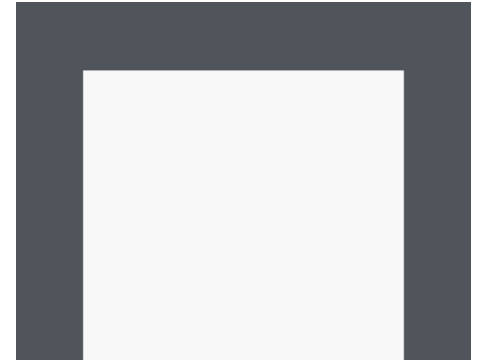
Rear



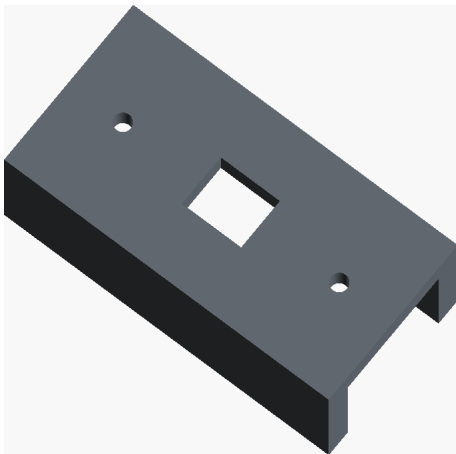
Low end



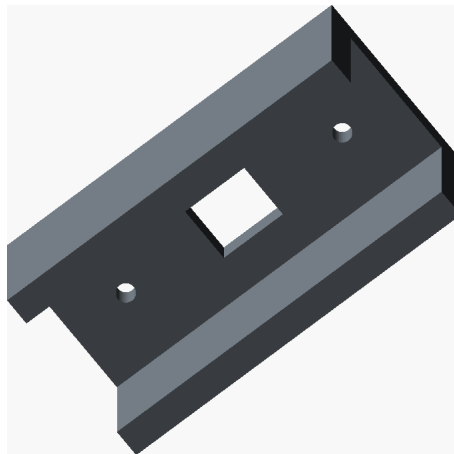
Tower end



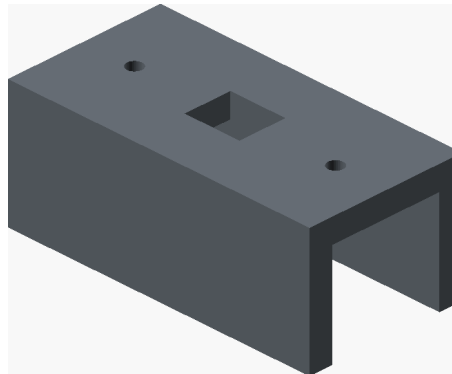
Top oblique



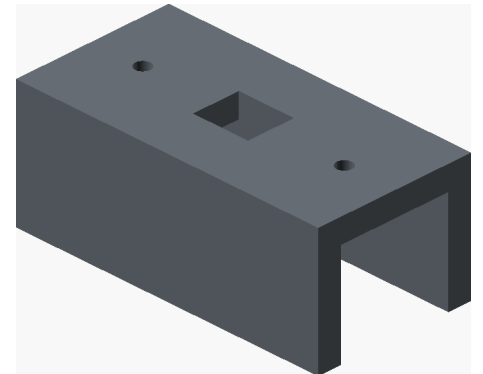
Bottom oblique



Front oblique



Rear oblique



r-white

Bounds: 70.49 x 157.94 x 21.66 mm. Nearest lettering face shown front/rear.

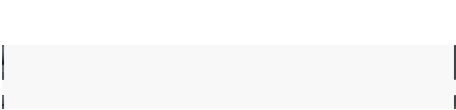
Front



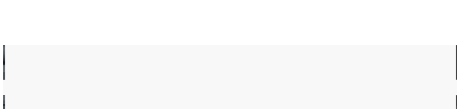
Rear



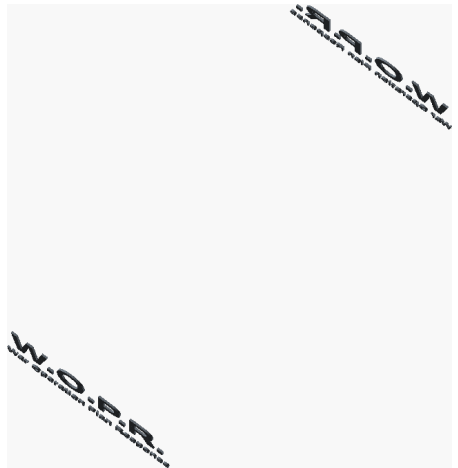
Low end



Tower end



Top oblique



Bottom oblique



Front oblique



Rear oblique



r-white

Bounds: 70.49 x 219.94 x 21.66 mm. Nearest lettering face shown front/rear.

Front

Rear

Low end

Tower end

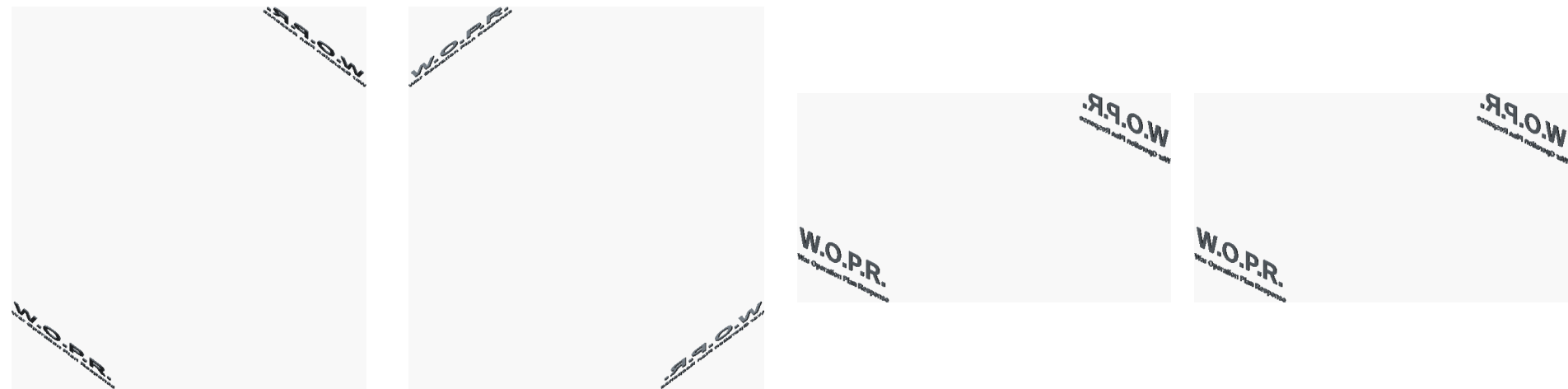


Top oblique

Bottom oblique

Front oblique

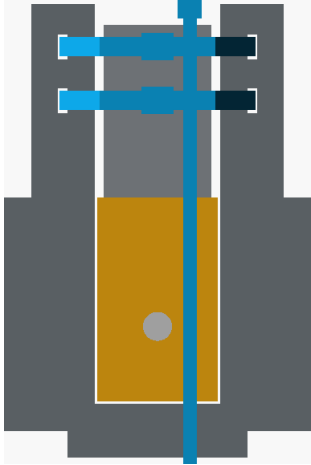
Rear oblique



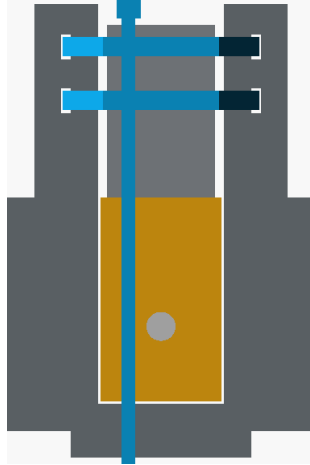
# Two side ties and one over the top

Blue: complete tie loops and heads. Gray: supports and motor can. Yellow: gearbox.

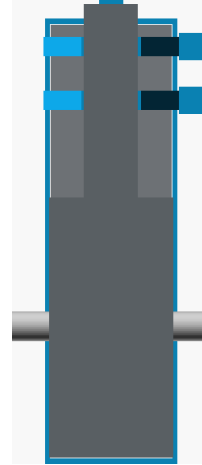
Front



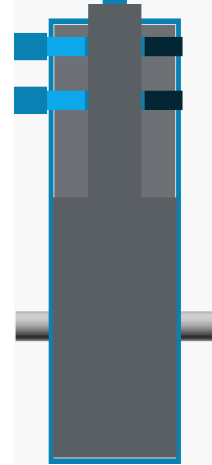
Rear



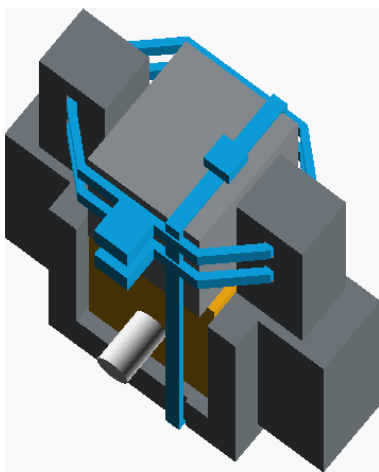
Low end



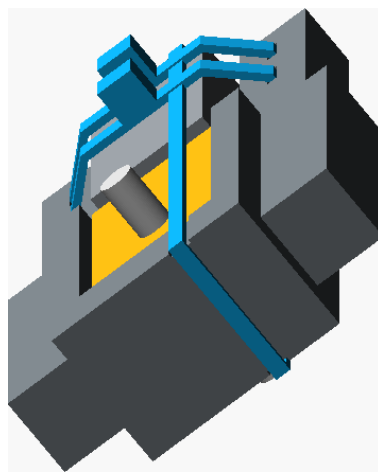
Tower end



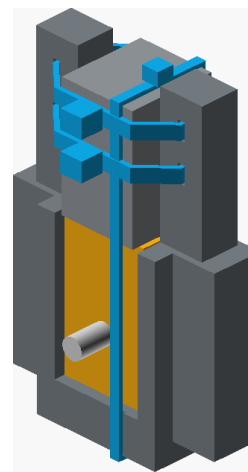
Top oblique



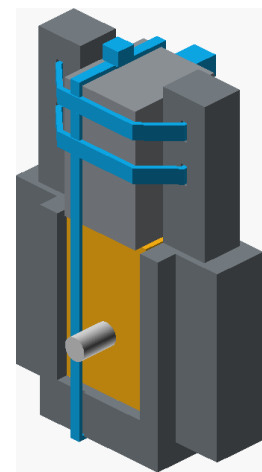
Bottom oblique



Front oblique



Rear oblique



# WOPR reinforced editions - current print kits

2026-09-29. SOLID BATTERY BLOCK / OUTER-WELL LIGHTING REVISION. These files replace the earlier thin-wall kits for new prints. Five editions are included: seven-module-per-side organizer, six-per-side organizer, wide low-axle eight-wheel robot, tucked four-wheel robot, and wide tucked eight-wheel robot. Earlier exports remain historical references. Do not mix old and reinforced mating parts.

## What is reinforced

Structural case walls, roof skins, lid floor and rim, cup walls/floors/dividers, LED retainer thickness, controller tray, motor retaining cheeks and wheel-cover skins are nominally 5 mm. All chassis/base plates are now 10 mm, including the organizers so their previous 6 mm base is not weakened. Robot motor bearing feet are at least 10 mm and are part of the chassis. There are no detachable motor-mount flanges or separate wheel-well covers.

Every robot chassis is one continuous printed solid: the main base, all four motor mounts and every wheel well are fused together. The standard version uses four removable 5 mm retaining caps. The second version uses three zip ties per motor and omits the caps. Two horizontal ties surround the motor; a third passes over its top and under its existing 10 mm bottom seat. The motor still inserts from above. Zip-only upper supports are now 10 mm deep instead of 23 mm and 11.8 mm wide instead of 12.7 mm. The side ties can turn inward against the motor faces. Their tunnels retain 5 mm walls on both sides and 5.4 mm between levels. Screw-cap motor supports are unchanged.

The wheel wells include 6.5 mm of axial fitting travel. Install bare motors from above, then raise each wheel through the underside opening at the service offset and slide it onto its shaft. On the eight-wheel models, the two inner tires at each end share a continuous arched chamber. This provides fitting room without a thin divider. Open shaft slots admit the motor's fixed double-ended shaft.

The shell has a continuous support ledge, larger base bosses and thicker lid supports. The gray material below the recessed lettering has extra backing so the lettering does not reduce its nominal 5 mm structural section. The white inlays remain decorative material volumes, not 5 mm structural parts. Screw bores, counterbores, wire passages, windows, service openings and assembly clearances are intentional local openings. Nominal wall thickness is not a claim that every fastener edge has 5 mm of material.

## Edition dimensions

All bodies remain 279.4 mm long and nominally 165 mm high. Organizers remain 155 mm wide. The four-wheel robot is 158 mm wide, giving 2.8 mm clearance between the unused inner shaft tips. BOTH eight-wheel bodies are now 220 mm wide. All robot side walls run continuously to the bottom edge; the wheel wells sit behind the 5 mm skins, with no wheel arches cut into the outer sides.

The low-axle eight-wheel chassis is 209.6 mm wide inside its 220 mm body. Its axles remain 22 mm below the body edge; total nominal height is 218.5 mm. The tucked versions keep axles 18 mm above the body edge, exposing 13.5 mm of tire and giving a nominal total height of 178.5 mm.

Thicker motor bases consume clearance below the case: the tucked robots have approximately 7.1 mm minimum ground clearance beneath the integrated motor feet; the low-axle robot also has approximately 7.1 mm. The third motor tie reduces the zip version minimum to approximately 5.9 mm. These are low indoor chassis. Tire projection below the body is not the same as ground clearance beneath the motor mounts.

The LCD opening remains 50.1 x 39.6 mm with the established mounting-hole pitch. Its bezel is now 5 mm thick; the glass and PCB seat move 3 mm inward. A new LCD fit coupon is included. LED PCB dimensions and module counts stay unchanged. All robots use six modules per side.

Organizer main cups are now 166.4 x 111 x 88 mm. Five-millimeter floors/dividers retain four 38.1 mm shallow cells and four 83 mm deep cells. Print either the shallow-left or shallow-right version. The tower cup is 58.4 x 94.4 x 100 mm, with four 95 mm deep cells. Its smaller footprint leaves stronger material around the roof opening.

## Files to print

Choose one folder under output/reinforced: seven, six, exposed, four or wide. Each contains a self-contained SCAD and printable STL files. shell-two-color.3mf contains the reinforced shell and aligned white inlays as one object; assign gray and white filament explicitly. It is a geometry project, not a printer profile.

For a single-color body, print r-shell.stl; its recessed lettering has the thicker backing. For two colors, use the 3MF or import r-shell and r-white together without independently centering them.

Every edition: one r-shell and two r-retainers. Organizers use r-frame. Robots use ONE base: r-frame for screw-cap retention, or r-frame-zip for zip-tie retention. Do not print both bases for one robot. Optional r-white supplies the inlay material. Print r-lcd-coupon before the full body.

Organizers: add ONE r-cup-left or r-cup-right, and one r-cup-tower. Robots: add one r-lid and one r-tray. The screw-cap base needs four r-caps. The zip-tie base needs no r-caps; supply eight 3.6 mm wide side ties (at least 250 mm long, at most 1.2 mm thick) and four 2.5 mm wide top ties (at least 250 mm long, at most 1.0 mm thick). There are no r-pod or r-hood parts in this revision.

Use the updated native Bambu project matching the robot and retention method. Its part list contains one complete chassis. The previous Bambu projects with separate motor pods or wheel hoods are superseded.

Wheel hub seating depth still needs a physical check. The nominal wheel inner face is 12.5 mm from the motor center; the service position moves it 6.5 mm away from the motor, providing 0.7 mm clearance beyond the nominal 18.3 mm shaft tip. Support the gearbox and never hammer a wheel onto a shaft.

## Material and orientation

Use PETG or ASA for structural parts, with the same material family for inlays. A starting point is 0.2 mm layers, five or more perimeters, six top/bottom layers and 35% gyroid infill. Use solid local infill at screw bosses, motor feet and braces. The CAD wall dimensions are real geometry; increasing slicer perimeter count alone does not create this reinforcement.

Print the shell upright. Print tray, retainers and chassis in orientations that give broad supported faces; inspect the tall chassis posts before slicing. The one-piece chassis needs careful support placement beneath the plate and inside the fixed wheel wells. Print motor caps with their roof on the plate and cheeks upward. Inspect and clear all motor pockets, counterbores and wheel openings.

Parts need about 280 x 155 mm, 280 x 158 mm or 280 x 220 mm of usable bed area plus brim, according to the edition. Do not scale parts to fit a smaller printer. No slicer toolpath or physical strength test has been completed for this revision.

## Fasteners

- Every edition: eight M3x20 base screws, with heads no taller than 3 mm.
- Robots: four M3x20 main-lid screws; four 3.6 mm tray ties at least 150 mm long. The tray no longer uses screws.
- Screw-cap robots only: eight M3x14 screws secure the four motor caps.

- Zip-tie robots only: eight 3.6 mm side ties and four 2.5 mm top ties, at least 250 mm long.
- Eight-wheel battery shelves: four 3.6 mm ties, at least 300 mm long.
- Center battery: two straps up to 10 mm wide and 2 mm thick, at least 350 mm long.
- Sensors: thirty-two M2x6 screws, four per board; verify actual screw penetration before tightening.
- LED retainers: eight M2x10 screws. LCD: four M2x6 screws; verify actual board thickness and safe tip depth with the coupon.

M3 pilots are 2.5 mm and are intended for plastic-compatible screws. Hand-tighten; the parts do not contain modeled heat-set inserts. Confirm screw lengths on the actual prints before tightening against electronics.

## Robot assembly order

1. Remove all support material from the one-piece chassis. Check the motor pockets, shaft slots, fixed wheel chambers and screw or zip-tie passages. Do not try to assemble the earlier loose motor-mount or wheel-cover parts onto this chassis.
  2. For the zip-tie version, each tie makes a complete horizontal loop around the motor and both side supports. Feed it front-to-back through the LEFT support tunnel, across the back of the motor, then back-to-front through the RIGHT support tunnel. Close the head across the front of the motor. Repeat at the second height. The 1.8 x 4.6 mm tunnels run beside the motor, never through its pocket. Leave the loops loose until the motor is seated. Put the side-tie heads toward the nearest outer side of the case, away from the battery shelves. Pre-thread the third tie under the motor foot and through its two vertical reliefs; this path is offset 6 mm from the shaft centerline.
  3. Lower each BARE motor from above, motor can upward. Its fixed double shaft enters the open U-slots and the gearbox seats on the integral 10 mm foot. Do not install the wheels before this step.
  4. Insert each wheel from below at its service position: 6.5 mm farther away from the motor than its running position. Raise it until the hub aligns with the shaft, then slide it inward and press it fully onto the shaft while supporting the gearbox. Repeat for one outward wheel per motor on the four-wheel model, or both shaft ends on either eight-wheel model.
  5. Verify every wheel turns freely. For screw-cap retention, fit the cap with a trimmed foam pad in the nominal 4 mm top gap, then install two M3x14 screws. For zip-tie retention, tighten the two side ties against the motor faces. Close the third tie over the motor top to hold it down onto the bottom seat. Keep the top band and head clear of electrical terminals; insulate nearby solder joints. No added top cap is required. Trim the tails and confirm the motor cannot lift or rock. Do not use a retaining cap on the zip-tie base.
  6. Route motor wires clear of tires and tie tails. Turn the selected USB pack across the case: its 104 mm dimension runs across the width, and its 52.3 mm dimension runs lengthwise. Seat it on the 8 mm rails above the LED leads and strap it through the base slots. Mount the controller and drivers on the insulating carrier. Secure the 5 mm tray with four ties: down one tray slot, through the horizontal tunnel below the post, up the paired tray slot, then over the tray.
  7. Fit LED modules with the reinforced retainers, then the LCD using the new fit coupon. Fit the motor-power switch. Attach the body with eight M3x20 screws and the lid with four M3x20 screws.
- For wheel service, remove the body and loosen the motor cap or ties as needed. Slide the wheel away from the motor into the service space before lowering it through the underside opening.

Organizer assembly: install display and RGB hardware first, fit the 10 mm base, then lower the two reinforced cups onto their support ledges. The cups remain removable for access.

## Level battery shelves and tray

The tray and both eight-wheel shelves have top surfaces at Z72.3 mm. Each platform is 56 x 53.12 mm: its across-case width is reduced by 20%. It is a continuous solid CAD block from the curved wheel-well roof to the platform top. Two 4.6 x 1.8 mm tunnels run horizontally through each block, beneath a 5 mm top skin. Feed each tie straight through a tunnel, then wrap it around the pack; keep its head on the upper outside face above the motor supports. Verified pack envelope per platform is 54 x 49.12 x 40 mm, including any padding; use protected battery packs. The native Bambu projects include two 100% rectilinear infill modifiers for the battery blocks. Keep these modifiers enabled to print the blocks densely. Other parts retain their existing settings. The blocks supplement the original central USB pack bay. The four-wheel model has no center wheel cavities and therefore no added wheel-cavity shelves. Do not connect separate packs together without a suitable power circuit.

## Lighting holes

All robot models now have 22 holes for 5 mm LEDs: eight at the OUTER wheel positions, six between the wheels (three per side), and eight below the tray in a 4 x 2 grid. The inner wheel wells have no LED holders or LED holes. All three also have FOUR 8 mm LED holes, one centered 20 mm below each front/rear sensor. These counts exclude the existing RGB modules.

The 5 mm holes are nominal lens bores. Wheel-roof bosses include 6 mm flange seats, 1 mm deep, so the lens stays recessed from the tire space. Use LED lens bodies no longer than 8.7 mm below the flange and flanges no wider than 6 mm; check actual parts before printing. Seven millimeters of lead-routing space is reserved above the base and wheel bosses. The 8 mm holes reserve a 9 mm diameter, 8 mm deep internal LED/wire space. Fit and insulate the leads, add strain relief, and confirm all lenses stay clear of rotating tires. Add appropriate current limiting; this revision adds mounting provisions, not LED control firmware.

## Eight distance-sensor mounts

Mount eight Adafruit 3967 VL53L1X boards inside the shell: two at the low-X front, two at the tower-end rear, and two on each long side. Each mount has four blind 1.6 mm M2 screw pilots on the manufacturer's 20.32 x 12.70 mm pitch. The central optical opening is 10 mm. The PCB face seats 8 mm behind the exterior surface, leaving component and connector space. Front/rear sensor centers are at Y40 and W-40, Z64; side centers are X85 and X170, Z64. Front/rear 8 mm LED holes are at Z44 directly below those centers.

Dimensions come from Adafruit's Eagle board: outline 25.4 x 17.78 mm; four 2.5 mm board holes at X2.54/22.86 and Y2.54/15.24; sensor centered at X12.7, Y8.89. The supplied firmware does not yet read these added sensors. Their common default I2C address is 0x29, so eight boards need separate address initialization via XSHUT or an I2C multiplexer before shared-bus use.

## Electronics and Wi-Fi

The original drive electronics and firmware remain unchanged: four Adafruit 3777 motors; Adafruit 3766 wheels; ESP32-DevKitC V4; two Adafruit 3297 DRV8833 drivers; Anker A1259 USB pack (104 x 52.3 x 26 mm). Retain the 5 V, 2.5 A fused supply and 0.5 A current limit per motor, set with four 0.40-ohm sense resistors. Use a 12 mm latching motor-power switch rated at least 3 A at 5 V DC, two 470 uF / 10 V driver capacitors and four 100 nF motor-terminal capacitors. The USB-C sink breakout must be rated 3 A and use 5.1k CC resistors; never select a higher-voltage PD trigger. Use 18-22 AWG power wiring, insulated connectors, two 10 mm battery straps at least 350 mm long and a nylon-mounted insulating controller carrier no larger than 90 x 65 mm. Reuse Adafruit 3315 LCD and the existing Seeed RGB modules.

The existing sketch in firmware/wopr-robot/wopr-robot.ino creates the WOPR access point and captive controller page. Join WOPR; use its Wi-Fi sign-in prompt, or open <http://192.168.4.1>. The public setup password remains wopr-car-setup and should be changed locally. Left driver: AIN1/AIN2 to GPIO25/26 (front motor), BIN1/BIN2 to GPIO27/14 (rear). Right driver: AIN1/AIN2 to GPIO32/33 (front), BIN1/BIN2 to GPIO18/19 (rear). Both SLP pins go to GPIO23 with a 10k pulldown; both open-drain FLT pins go to GPIO21 with a 10k pullup to 3.3 V. Common ground; never connect driver outputs together. The unchanged sketch was previously compiled with Espressif Arduino core 3.3.8; this task does not flash hardware.

Test direction, release-to-stop, STOP, Wi-Fi loss, driver fault and the manual switch with wheels raised. Disconnect the battery before attaching PC USB for programming. Start driving tests with RGB/LCD power disconnected. The lighting current must be measured and brightness limited so the complete system stays below 2.5 A continuous. The drive sketch does not regulate lighting.

## Verification and limits

The builder checks watertight connected structural meshes, actual exported wall/base/foot thicknesses, tray tie paths, shelf pack volumes, sensor/connector envelopes and LED lead space, wheel and battery clearance, LCD/LED fit volumes, and bare-motor insertion, wheel lifting and axial fitting paths. The zip-tie check now includes the complete external belt and lock head against the motor, chassis, wheels and enclosure. The prior straight-through slots are superseded; their earlier empty-slot check did not prove a usable loop. White lettering consists of separate glyph solids by design.

This is a stronger CAD revision, not a certified load or drop rating. Print fit, motor torque under the heavier case, steering scrub, operating temperature, runtime and impact durability remain untested. Plastic TT gears and press-fit wheel hubs remain mechanical limits. Check an actual motor and wheel against the fit geometry before relying on the full set. Zip-tie retention also needs a physical pull/rocking test and inspection after driving.

## Component references

<https://www.adafruit.com/product/3777>

<https://www.adafruit.com/product/3766>

<https://www.adafruit.com/product/3297>

<https://www.adafruit.com/product/3315>

<https://www.anker.com/nz/products/a1259-built-in-cable-power-bank-10000mah>

[https://docs.espressif.com/projects/esp-dev-kits/en/latest/esp32/esp32-devkitc/user\\_guide.html](https://docs.espressif.com/projects/esp-dev-kits/en/latest/esp32/esp32-devkitc/user_guide.html)

<https://www.adafruit.com/product/3967>

<https://github.com/adafruit/Adafruit-VL53L1X-PCB>